ASU
جامعة العلوم التطبيقية
APPLIED SCIENCE UNIVERSITY

Basira

Intelligent Decision Support System with Predictive Analytics for Small Businesses

Graduation Project 2 - Chapters 1 to 5

by

Khaled Waled Abdulakder - 20236110155

SUMMER SEMESTER 2025/2026

Supervisor

Dr.Hilal Al-Bayatti

Table of Contents

Chapter 1: Introduction 1

1.1 Overview 1

1.2 Project Motivation 2

1.3 Problem Statement 2

1.4 Aim and Objectives 3

1.5 Scope 3

1.6 General Constraints 4

1.7 Document Organization 5

Chapter 2: Background and Related Works 6

2.1 Theoretical Background 6

2.1.1 Decision Support Systems 6

2.1.2 Business Intelligence 6

2.1.3 Small Business Data Management 7

2.1.4 Key Performance Indicators 7

2.1.5 Data Visualization and Dashboards 8

2.1.6 Predictive Analytics 8

2.1.7 Natural Language Processing 9

2.1.8 Conversational AI Business Assistant 9

2.1.9 Automated Reports 10

2.1.10 Smart Notifications and Alerts 10

2.2 Analysis and Limitation of Existing Systems 10

2.2.1 Microsoft Power BI 11

2.2.2 Zoho Analytics 11

2.2.3 QuickBooks Online 12

2.2.4 Oracle NetSuite 12

2.2.5 Spreadsheet-Based Systems 13

2.2.6 Comparison of Existing Systems 13

2.3 Need for New System 15

Table of Contents (continued)

Chapter 3: Project Methodology 16
3.1 Project Planning 16
3.1.1 Feasibility Study 16
3.1.2 Gantt Chart 18
3.2 Software Development Methodology 20
3.2.1 Agile Software Development Methodology 20
3.2.2 Phases of Software Development Methodology 21
3.3 Analysis of the New System 22
3.3.1 User Requirements 22
3.3.2 System Requirements 26
3.3.3 Domain Requirements 27

Chapter 4: Software Design 28
4.1 Use Case Diagram 28
4.2 Design of Database 30
4.3 Class Diagram 33
4.4 Sequence Diagram 35
4.5 Activity Diagram 36

Chapter 5: Software Implementation 38
5.1 Software and Hardware 38
5.1.1 Software 38
5.1.2 Hardware 39
5.2 Implementation 39
5.2.1 Client and Server Architecture 40
5.2.2 Object Relational Mapping 41
5.2.3 Refinements to the GP1 Design 42
5.2.4 User Authentication 43
5.2.5 Business Profiles 44

Table of Contents (continued)

5.2.6 Product and Service Catalogue 44
5.2.7 Sales and Expense Records 45
5.2.8 Customer Visit Metrics 46
5.2.9 CSV and Excel Imports 46
5.2.10 Dashboard and Key Performance Indicators 48
5.2.11 Predictive Analytics 50
5.2.12 Customer Feedback Analysis 52
5.2.13 AI Business Assistant 53
5.2.14 Alerts and Notifications 54
5.2.15 Report Generation 55
5.2.16 Administration Panel 56
5.2.17 Security and Error Handling 56
5.2.18 Local Execution and Deployment Preparation 57
5.3 Chapter Summary 57

References 58

List of Figures

Figure 3.2: Agile Software Development Lifecycle Diagram. 21
Figure 4.1: Use Case Diagram for the Intelligent Decision Support System. 28
Figure 4.2: Entity Relationship Diagram / Database Design. 31
Figure 4.3: Class Diagram for the proposed system. 34
Figure 4.4: Sequence Diagram for AI Business Assistant Request. 35
Figure 4.5: Activity Diagram for the proposed system. 36
Figure 5.1: Implemented client-server communication and external services. 40
Figure 5.2: Basira sign-in interface with password and Google options. 43
Figure 5.3: Separate business profiles in the demonstration account. 44
Figure 5.4: Expense-entry form with amount, category and date. 45
Figure 5.5: Expense import preview excerpt; six rows are valid and none has been confirmed. 47
Figure 5.6: Dashboard KPI cards for the synthetic cafe, 18 July to 15 October 2026. 48
Figure 5.7: Recorded revenue, all expenses and estimated profit over time. 49
Figure 5.8: Cost and customer-record coverage displayed beside the dashboard. 49
Figure 5.9: Forecast history review with recorded-date and gap counts. 50
Figure 5.10: Saved forecast interface excerpt using a synthetic automated-test series. 51
Figure 5.11: Feedback insights excerpt from an automated-test fixture with simulated classifications. 52
Figure 5.12: Assistant reply and saved context excerpt; reply supplied by an automated-test provider. 53
Figure 5.13: Alert-check excerpt for the synthetic cafe, including a cost-coverage warning. 54
Figure 5.14: Saved financial report for the synthetic cafe, 1 to 30 September 2026. 55
Figure 5.15: Administration statistics from an automated-test fixture. 56

List of Tables

Table 2.1: Comparison of existing systems and the proposed system. 14
Table 3.1: Project Schedule. 19
Table 3.2: User Requirements. 23
Table 3.3: Functional Requirements. 24
Table 3.4: Non-Functional Requirements. 25
Table 3.5: Hardware Requirements. 26
Table 3.6: Software Requirements. 26
Table 4.3: Database Entities Description. 31
Table 5.1: Software used in the implementation. 38
Table 5.2: Hardware and execution environment. 39
Table 5.3: Main implemented model relationships. 41
Table 5.4: Principal refinements from design to implementation. 42
Table 5.5: Supported import data and principal validation rules. 46
Table 5.6: Forecast horizons and implemented history requirements. 50
Table 5.7: Default alert rules and data safeguards. 54

Chapter one

Introduction

1.1 Overview

Small businesses play an important role in the economy by providing products, services, employment opportunities, and local market competition. However, many small business owners and managers face difficulties when they need to make decisions based on business data. These decisions may include identifying profitable products, controlling expenses, understanding customer feedback, predicting sales, and planning future growth. In many cases, small businesses do not have dedicated business analysts, expensive business intelligence tools, or enough background in finance and management to analyze their performance effectively.

An Intelligent Decision Support System with Predictive Analytics for Small Businesses is a web-based platform designed to help small business owners understand their business performance in a simple and clear way. The system allows users to enter business data manually or import structured data files such as CSV and Excel files. This data may include sales records, expenses, customer numbers, inventory costs, and customer feedback. After the data is entered into the system, it is organized, analyzed, and displayed through interactive dashboards and visual reports. [1]

The proposed system will use artificial intelligence and data analysis techniques to transform business data into useful insights. Predictive analytics will be used to forecast future sales, estimate expenses, and identify trends in business performance. Natural Language Processing (NLP) will be used to analyze customer feedback and support an AI business assistant that can answer user

questions in simple language. This makes the platform suitable for small business owners who do not have a strong business or technical background. [2], [3]

The system will also include smart notifications and automated reports. These features will help users identify important changes, such as a decrease in sales, an increase in expenses, or negative customer feedback. Instead of requiring the user to manually study complex tables and calculations, the system will present results in a user-friendly form using charts, summaries, recommendations, and alerts.

1.2 Project Motivation

The main motivation behind this project is the real need for simple and affordable decision-support tools for small businesses. Many small businesses have useful data, but they do not use it effectively. For example, a small restaurant may have daily sales records, customer comments, and expense details, but the owner may not know how to analyze this information to improve business performance. A small shop may know how much it sells every day but may not clearly understand which products are more profitable or when sales are expected to decrease.

Large companies usually have access to professional analysts, advanced business intelligence systems, and financial consultants. Small businesses often cannot afford these services. As a result, many decisions are made based on guessing, personal experience, or incomplete information. This can lead to poor planning, unnecessary expenses, weak customer satisfaction, and missed growth opportunities.

This project was selected because it combines computer science with business and management. It applies modern web development, databases, artificial intelligence, predictive analytics, and data visualization to solve a practical business problem. The system is not only a dashboard; it is designed to act as a simple digital advisor for users who may not fully understand business terminology. The goal is to make business analysis easier, faster, and more accessible for small businesses.

Another motivation is the increasing importance of artificial intelligence in business systems. AI can support decision-making by finding patterns in data, generating summaries, predicting future results, and explaining business performance in simple terms. By integrating an AI assistant into the platform, the project becomes more useful for non-expert users because they can ask questions such as “Why did my profit decrease this month?” or “What can I do to improve sales?” and receive understandable advice.

1.3 Problem Statement

Many small businesses struggle to make effective business decisions because they lack the tools, knowledge, and resources needed to analyze their data. Although business owners may collect sales, expenses, customer feedback, and inventory information, this data is often stored in notebooks, spreadsheets, receipts, or separate systems. This makes it difficult to understand the current performance of the business or predict future problems.

The main problem is that existing advanced business management and analytics systems are often expensive, complicated, or designed for users with business and technical experience. Small business owners who do not have a business background may find it difficult to understand complex reports, financial indicators, forecasting models, or business intelligence dashboards. Hiring consultants or analysts can also be costly, especially for startups and small businesses with limited budgets.

Because of this problem, small businesses may face several issues. They may not recognize that expenses are increasing until it is too late. They may fail to identify their best-selling products or services. They may ignore customer feedback because it is difficult to analyze manually. They may also be unable to predict future sales, which can affect inventory planning, budgeting, and growth decisions.

Therefore, there is a need for an intelligent, affordable, and user-friendly decision support system that helps small businesses manage data, understand business performance, receive AI-generated recommendations, and make better decisions without needing advanced business knowledge.

1.4 Aim and Objectives

The aim of this project is to develop an Intelligent Decision Support System with Predictive Analytics for Small Businesses. The system will help small business owners and managers analyze their business data, understand performance, predict future trends, and receive simple AI-supported recommendations through an easy-to-use web platform. [1], [2], [3]

The objectives of the proposed system are:

1. To develop a scalable web platform that allows small businesses to manage operational data through manual entry and CSV/Excel file import.
2. To design interactive dashboards that display important business Key Performance Indicators (KPIs), such as revenue, expenses, profit margin, customer numbers, and sales trends.
3. To implement a predictive analytics module that uses time-series forecasting to predict future sales, estimate expenses, and identify business performance trends. [2]
4. To integrate an NLP-powered AI business assistant that can analyze customer feedback and provide simple, understandable business advice for users without a strong business background. [3]
5. To automate the generation of periodic business reports that summarize business performance in a clear and organized format.
6. To develop a threshold-based alert system that notifies users about important business changes, such as unusual expense increases, sales drops, or negative feedback patterns.
7. To design a user-friendly interface that makes the system easy to use for small business owners, managers, and startup founders.

1.5 Scope

The scope of this project is to develop a web-based intelligent decision support platform for small businesses. The system will focus on helping users enter, manage, analyze, and understand business data. The platform will include modules for business profile management, data entry, file-based data import, dashboards, predictive analytics, customer feedback analysis, AI assistant interaction, report generation, and smart notifications.

The system will support common small business data, such as sales records, expenses, customer feedback, customer numbers, product or service categories, and basic inventory or cost information. The system will not be limited to one specific business type, but it will be designed mainly for small

businesses such as small shops, cafés, restaurants, service providers, and startups.

The system will include the following main functions:

The user can create and manage a business profile.

The user can manually enter business data.

The user can upload structured CSV or Excel files containing business records.

The system can display business performance using charts and dashboards.

The system can generate forecasts for sales and expenses.

The system can analyze customer feedback using NLP techniques.

The user can interact with an AI business assistant through a chat interface.

The system can generate business reports.

The system can send smart alerts when important changes or anomalies are detected.

The project will focus on the analysis, design, and development of the main features required for a working prototype. Advanced accounting functions, full enterprise resource planning, payroll management, payment processing, and legal financial auditing are outside the scope of this project. The system will not replace professional accountants or business consultants, but it will provide basic decision-support guidance that helps small business owners understand their situation and make more informed decisions.

1.6 General Constraints

Several constraints may affect the development of this project. The first constraint is time. Graduation Project 1 focuses mainly on proposal, research, analysis, and design, while Graduation Project 2 will focus more on implementation and testing. Because the project includes multiple modules, such as dashboards, forecasting, AI assistant integration, file parsing, and notifications, the available time must be managed carefully.

The second constraint is data availability. Predictive analytics requires suitable historical business data to generate useful forecasts. Since real small business data may be difficult to obtain due to privacy and confidentiality, sample datasets may be used during development and testing. The accuracy of forecasts depends on the quality, size, and consistency of the data entered into the system. [2]

The third constraint is AI integration. The system will rely on an external AI service, such as Google AI API, to provide AI assistant features and natural language analysis. This creates a dependency on API availability, usage limits, cost, response quality, and internet connection. The AI responses must also be designed carefully so that the advice is simple, helpful, and not misleading. [3]

The fourth constraint is user background. The target users may not have strong business or technical knowledge. Therefore, the system interface, reports, alerts, and recommendations must be simple and easy to understand. This increases the importance of good user interface design and clear wording.

The fifth constraint is project complexity. Combining web development, database design, dashboard visualization, forecasting algorithms, file processing, and AI integration requires careful planning. To keep the project achievable, the most important features will be implemented first, and additional advanced features may be considered as future work.

Despite these constraints, the project can be completed successfully by focusing on the main requirements, using suitable technologies, and developing the system in stages.

1.7 Document Organization

This project report is organized into seven chapters. Each chapter focuses on a specific part of the project.

Chapter 1: Introduction

The first chapter gives a general overview of the project. It explains the project idea, project motivation, problem statement, aim and objectives, scope, and general constraints. This chapter helps the reader understand the purpose of the Intelligent Decision Support System with Predictive Analytics for Small Businesses.

Chapter 2: Background and Related Works

The second chapter discusses the theoretical background related to the project. It explains important concepts such as decision support systems, business intelligence, predictive analytics, dashboards, data visualization, Natural Language Processing, and AI business assistants. It also analyzes existing systems and discusses their limitations. Finally, it explains the need for the proposed system.

Chapter 3: Project Methodology

The third chapter will explain the project planning and methodology. It will include the feasibility study, project schedule, software development methodology, and analysis of the new system. It will also describe the functional requirements, non-functional requirements, system requirements, and domain requirements.

Chapter 4: Software Design

The fourth chapter will focus on the software design of the proposed system. It will include diagrams such as use case diagram, database design, class diagram, sequence diagram, and activity diagram. These diagrams will explain how the system components interact with each other.

Chapter 5: Implementation

The fifth chapter will explain the implementation of the system. It will describe the software and hardware tools used in the project. It will also explain the main system interfaces and modules, such as dashboard, data entry, forecasting, AI assistant, reports, and alerts.

Chapter 6: Testing

The sixth chapter will focus on testing the system. It will include testing types, testing scenarios, expected results, actual results, and discussion. The purpose of this chapter is to ensure that the system works correctly and meets the intended requirements.

Chapter 7: Conclusion and Future Work

The final chapter will summarize the project and explain the main achievements. It will also suggest future improvements that can be added to the system, such as advanced financial analysis, integration with accounting systems, mobile application support, and more accurate forecasting models.

Chapter Two
Background and Related Works

2.1 Theoretical Background

This chapter discusses the theoretical background of the proposed project and analyzes related systems. Since the project is an Intelligent Decision Support System with Predictive Analytics for Small Businesses, it is important to explain the main concepts related to decision support systems, business intelligence, predictive analytics, dashboards, Natural Language Processing, AI assistants, and small business data management.

2.1.1 Decision Support Systems

A Decision Support System (DSS) is a computer-based information system that helps users make decisions by collecting, organizing, analyzing, and presenting data in a useful form. Unlike ordinary data storage systems, a DSS does not only store information. It supports the user by providing analysis, comparisons, reports, predictions, and recommendations. This makes it useful in situations where users need to make decisions based on different types of information. [1]

In business, decision support systems are commonly used to help managers understand performance, solve problems, and choose suitable actions. For example, a business owner may use a DSS to compare monthly sales, identify products with low profit, check customer feedback, and decide whether to increase marketing or reduce expenses. A DSS can also help the user understand the possible result of a decision by using forecasting and analysis. [1]

Decision support systems are especially useful for small businesses because many small business owners make daily decisions without formal analysis. These decisions may include how much inventory to buy, which products to promote, when to reduce expenses, and how to respond to customer complaints. By using a DSS, these decisions can become more data-driven and less dependent on guessing. [1]

The proposed system applies the concept of DSS by giving small business users a platform that collects business data, analyzes it, visualizes it, predicts future performance, and provides recommendations. The system will not make decisions instead of the user, but it will support the user by presenting clear information and AI-generated suggestions.

2.1.2 Business Intelligence

Business Intelligence (BI) refers to the tools, technologies, and processes used to transform raw business data into meaningful information. BI systems usually include data collection, data cleaning, reporting, dashboards, charts, and analysis. The purpose of BI is to help organizations understand what is happening in their business and make better decisions. [4], [6]

For example, a business may collect sales data every day. In its raw form, this data may only be a long list of transactions. When business intelligence tools are applied, the same data can show useful information, such as total monthly revenue, best-selling products, average order value, customer growth, and profit trends. This makes it easier for the business owner to understand the current condition of the business.

Business intelligence is important because data alone does not create value unless it is analyzed and understood. Small business owners may already have useful data in spreadsheets, receipts, sales

systems, or customer messages. However, without suitable tools, it can be difficult to identify patterns or make decisions based on that data.
The proposed system uses business intelligence concepts by providing dashboards, charts, summaries, and reports. These features will help users monitor Key Performance Indicators (KPIs) such as revenue, expenses, profit margin, customer numbers, and sales trends. Since the system targets users without a strong business background, the dashboards must be simple, visual, and easy to understand.

2.1.3 Small Business Data Management

Small business data management refers to the process of collecting, storing, organizing, and maintaining business-related data. This data may include sales records, expenses, product details, customer information, inventory costs, and customer feedback. Proper data management is necessary before any analysis or forecasting can be performed.
Many small businesses still manage their information manually or through separate spreadsheets. Although spreadsheets are useful, they can become difficult to manage when the amount of data increases. Errors may occur because of repeated manual entry, missing values, inconsistent formats, or incorrect formulas. For example, one file may write the date as day/month/year, while another file may use month/day/year. These small issues can affect the accuracy of the analysis.
A database can solve many of these problems by storing data in structured tables and connecting related records together. For example, sales records can be linked to products, expenses can be linked to categories, and feedback can be linked to customers or dates. This structure makes the data easier to query, analyze, and display.
In the proposed system, PostgreSQL will be used as the database, and Prisma ORM will help manage communication between the application and the database. Users will be able to enter data manually or import CSV/Excel files. The system will then organize the data and prepare it for dashboards, forecasting, reports, and AI analysis.

2.1.4 Key Performance Indicators

Key Performance Indicators (KPIs) are measurable values that show how well a business is performing. They help business owners focus on important performance areas instead of looking at all data at once. Common KPIs for small businesses include total revenue, total expenses, profit margin, sales growth, number of customers, average order value, and customer satisfaction.
KPIs are important because they simplify business performance. For example, instead of reading hundreds of sales records, the business owner can look at monthly revenue growth. Instead of reading every expense entry, the owner can view expense breakdowns by category. This makes decision-making faster and clearer.
In the proposed system, KPIs will be displayed in dashboards using charts and summary cards. These KPIs will help users answer questions such as: [6], [8]
Is the business making more sales this month than last month?
Which expense category is increasing?
Is the profit margin improving or decreasing?
Are customer numbers increasing?
Is customer feedback mostly positive or negative?
The system will also use KPIs as part of the alert system. For example, if expenses increase above a specific threshold or sales drop below a certain level, the system can notify the user.

2.1.5 Data Visualization and Dashboards

Data visualization is the process of representing data using visual elements such as charts, graphs, tables, and diagrams. It helps users understand data more quickly than reading raw numbers. Dashboards are interfaces that combine multiple visualizations and indicators in one place. [6], [8] Dashboards are useful in business because they provide a quick overview of performance. For small business owners, a dashboard can show sales, expenses, profits, and customer feedback on one screen. This saves time and helps the user identify important changes quickly. [6]

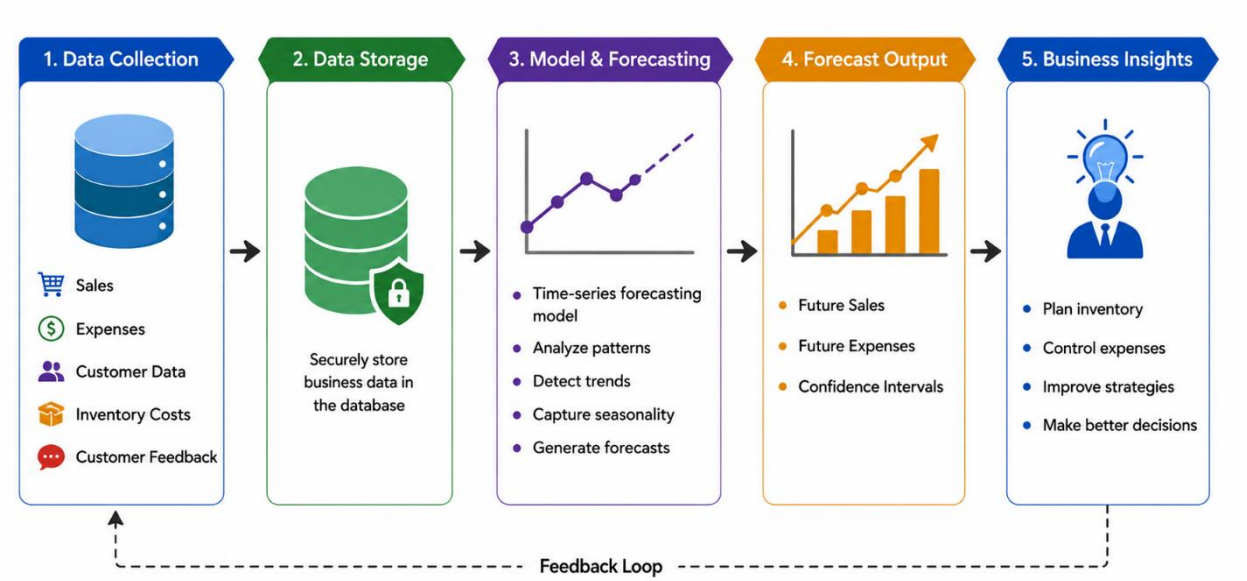
The proposed system will use dashboard visualizations to simplify business analysis. For example, a line chart can show sales trends over time, a bar chart can compare expense categories, and a pie chart can show revenue distribution by product or service. The dashboard should be designed to be simple and readable because the target users may not have experience with advanced analytics. Charts and dashboards also support the goal of making the system suitable for non-expert users. Instead of showing complex calculations, the system will display meaningful visuals supported by short explanations and AI-generated recommendations.

2.1.6 Predictive Analytics

Predictive analytics is the process of using historical data, statistical methods, and machine learning techniques to predict future outcomes. In business, predictive analytics can be used to forecast sales, estimate future expenses, identify seasonal trends, and detect unusual changes in performance. [2] For example, if a small shop has sales data for the last twelve months, predictive analytics can help estimate sales for the next month. If expenses have been increasing over time, the system can warn the user that future costs may become a problem. This helps business owners plan ahead instead of only reacting after problems occur. [2]

One common method in predictive analytics is time-series forecasting. A time series is a set of data points collected over time, such as daily sales, weekly expenses, or monthly revenue. Forecasting algorithms can analyze patterns in this data and generate predictions. These patterns may include growth, decline, seasonality, or repeated changes. [2]

The proposed system will use predictive analytics to forecast future business performance. The forecasting module will help users understand what may happen in the future based on past data. However, predictions are not guaranteed to be fully accurate. The quality of the prediction depends on the quality and amount of data available. Therefore, the system should present forecasts as decision-support information, not as certain results. [2]



2.1.7 Natural Language Processing

Natural Language Processing (NLP) is a field of artificial intelligence that focuses on enabling computers to understand, analyze, and generate human language. NLP can be used in many applications, such as chatbots, translation systems, text classification, sentiment analysis, summarization, and question answering. [3]

In the context of small businesses, NLP is useful for analyzing customer feedback. Customers may write comments, reviews, complaints, or suggestions in natural language. Manually reading and understanding all feedback can take time, especially when the business receives many comments. NLP can help by identifying whether feedback is positive, negative, or neutral. It can also extract common topics, such as price, product quality, delivery, customer service, or waiting time. [3]

The proposed system will use NLP to analyze customer feedback and generate simple insights. For example, if many customers complain about slow service, the system can detect this repeated issue and show it to the business owner. This allows the business to respond faster and improve customer satisfaction.

NLP will also support the AI business assistant. The assistant will allow users to ask questions in normal language and receive understandable answers. This is important because the target users may not know technical or business terms.

2.1.8 Conversational AI Business Assistant

A conversational AI assistant is a system that allows users to interact with software through a chat-based interface. Instead of clicking through many menus or reading complex reports, users can ask questions and receive responses in natural language. [3]

For small businesses, a conversational AI assistant can be very useful because it can explain business data in simple terms. For example, the user may ask:

What was my best-selling product this month?

Why did my expenses increase?

How can I improve my profit?

What does the forecast say about next month?

What are the main complaints from customers?
The AI assistant can analyze available data, summarize results, and provide recommendations. This makes the system more accessible for users without a business background. However, the assistant should be designed carefully. It should avoid giving unrealistic guarantees and should explain that recommendations are based on the available data.
In the proposed system, the AI assistant will be powered through an AI API. It will use the business data, dashboard results, forecasts, and feedback analysis to provide simple advice. This feature is one of the main differences between the proposed system and ordinary dashboards.
[3]

2.1.9 Automated Reports

Automated reports are generated documents or summaries that explain business performance over a specific period, such as daily, weekly, or monthly. Reports can include sales totals, expenses, profit, customer feedback, forecasts, and recommendations. [6]
Reports are important because business owners may not always have time to check dashboards. A report can summarize the most important information in a structured way. For example, a monthly report may show total revenue, total expenses, net profit, best-selling products, sales forecast, and customer feedback summary. [6]
The proposed system will generate periodic business reports automatically. These reports will help users track performance over time and keep records of business progress. The reports should be written in simple language and supported by charts where needed.

2.1.10 Smart Notifications and Alerts

Smart notifications and alerts are messages that inform users when something important happens in the system. In business decision support, alerts can help users respond quickly to problems or opportunities. [1]
For example, the system can notify the user when:
Sales decrease below a normal level.
Expenses increase above a selected threshold.
Profit margin becomes too low.
Customer feedback becomes mostly negative.
Forecasted sales are lower than expected.
A product or service performs better than usual.
These alerts are useful because small business owners may not have time to monitor all data manually. The system can automatically check important indicators and notify the user when attention is needed. In the proposed system, alerts will be threshold-based, meaning that the user or system can define specific limits that trigger notifications.

2.2 Analysis and Limitation of Existing Systems

To understand the need for the proposed system, several existing tools and systems can be analyzed. Many tools already help businesses manage data, create reports, or track finances. However, not all of them are suitable for small businesses with limited budgets and users without strong business knowledge.

2.2.1 Microsoft Power BI

Microsoft Power BI is a business intelligence platform that allows users to connect data sources, create dashboards, build reports, and analyze business performance. It is a powerful tool for organizations that need advanced data visualization and reporting. Users can connect Power BI to Excel files, databases, cloud services, and other data sources. [4]
Features of Microsoft Power BI:
It supports interactive dashboards and reports. [4]
It can connect to many data sources.
It provides strong data visualization tools.
It supports advanced business intelligence and analytics.
It can be used by businesses of different sizes.
Limitations of Microsoft Power BI:
Power BI may be difficult for non-expert users because building proper dashboards requires knowledge of data modeling, relationships, measures, and reporting concepts.
The user may need to clean and prepare data before using it effectively.
Advanced use may require learning DAX formulas and Power Query.
It mainly focuses on visualization and analysis, but it does not act as a simple business advisor for users without business knowledge.
Sharing and collaboration may require paid licenses. [4]
Power BI is a strong business intelligence tool, but it may not be the easiest solution for small business owners who want simple explanations, automatic advice, and predictive recommendations without technical setup.

2.2.2 Zoho Analytics

Zoho Analytics is a business intelligence and analytics platform that helps users create reports and dashboards from different data sources. It supports data import, visualization, and analysis. Zoho Analytics is useful for organizations that want cloud-based reporting and dashboard features. [5], [6]
Features of Zoho Analytics:
It provides dashboards and reports. [6]
It supports data import from different sources.
It includes business analytics tools.
It can be used for sales, marketing, finance, and operations analysis.
It offers cloud-based access.
Limitations of Zoho Analytics:
The system may still require users to understand data organization and analytics concepts.
Some advanced features may require paid plans. [5]
Small business users may need time to learn how to create useful dashboards.
It is a general analytics platform and may not be specifically designed for users without a business background.
AI advice and predictive recommendations may not be presented in a simplified way for every small business case.
Zoho Analytics is useful for reporting and analysis, but the proposed system aims to focus more directly on small businesses and provide simpler guidance through an AI business assistant.

2.2.3 QuickBooks Online

QuickBooks Online is an accounting and financial management system used by many small businesses. It helps users manage invoices, expenses, income, receipts, reports, and financial records. It is especially useful for accounting and bookkeeping. [7]
Features of QuickBooks Online:
It supports income and expense tracking. [7]
It allows users to create invoices and estimates.
It provides financial reports.
It can help with accounting and bookkeeping tasks.
It supports integrations with other business tools.
Limitations of QuickBooks Online:
The system focuses mainly on accounting rather than general decision support.
Users may still need accounting knowledge to fully understand financial reports.
Predictive analytics and AI business advice may be limited or not focused on simple business guidance.
It may include features that are not needed by very small businesses.
Some plans can be expensive for small businesses with limited budgets. [7]
QuickBooks is strong for financial management, but the proposed system focuses more on combining business dashboards, forecasting, feedback analysis, and conversational AI advice in one simple platform.

2.2.4 Oracle NetSuite

Oracle NetSuite is a cloud-based enterprise resource planning system that supports business management, financials, dashboards, reporting, and analytics. It is used by organizations that need integrated business management across departments. [8]
Features of Oracle NetSuite:
It provides enterprise resource planning features.
It supports dashboards and business analytics. [8]
It can manage financials, operations, inventory, and customer data.
It provides real-time business visibility. [8]
It is suitable for growing and larger organizations.
Limitations of Oracle NetSuite:
It may be too advanced and complex for small businesses.
Implementation can require professional setup and training.
The cost may be high for startups and small businesses. [8]
Many features may be unnecessary for small businesses that only need simple analysis and recommendations.
It is not mainly designed for non-expert users who need simplified business advice.
NetSuite is powerful for larger businesses, but the proposed system aims to provide a simpler and more focused solution for small businesses.

2.2.5 Spreadsheet-Based Systems

Many small businesses use Microsoft Excel, Google Sheets, or similar spreadsheet tools to record sales, expenses, and customer data. Spreadsheets are popular because they are flexible, familiar, and affordable.

Features of spreadsheet-based systems:

They are easy to start using.
They are low-cost or free.
They allow manual data entry.
They can perform calculations and create simple charts.
They are suitable for small amounts of data.

Limitations of spreadsheet-based systems:

Data can become difficult to manage as the business grows.
Manual entry may cause errors.
Files can be duplicated, lost, or edited incorrectly.
Forecasting and analysis require formulas or advanced knowledge.
Customer feedback analysis is difficult to perform manually.
Spreadsheets do not provide automatic AI advice or smart alerts.

Although spreadsheets are useful for simple record keeping, they are not enough for small businesses that need automated analysis, forecasting, and decision support.

2.2.6 Comparison of Existing Systems

Table 2.1 shows a comparison between existing systems and the proposed system.

System	Main Purpose	Strengths	Limitations	Suitability for Small Non-Expert Businesses
Microsoft Power BI [4]	**Business intelligence and dashboards**	**Strong visualizations, many data connections, advanced analytics**	**Requires technical knowledge and may be complex for non-expert users**	**Medium**
Zoho Analytics [5], [6]	**Cloud analytics and reports**	**Dashboards, data import, and reporting tools**	**Requires learning and setup, and advanced features may require payment**	**Medium**
QuickBooks Online [7]	**Accounting and financial management**	**Income tracking, expenses, invoices, and financial reports**	**Focuses mainly on accounting and may require accounting knowledge**	**Medium**
Oracle NetSuite [8]	**Enterprise resource planning**	**Complete business management and real-time analytics**	**Complex and expensive for small businesses**	**Low**
Spreadsheets	**Manual data recording and calculations**	**Flexible, affordable, and familiar**	**Error-prone, limited automation, and weak AI support**	**Medium**
Proposed Intelligent Decision Support System	**AI-supported decision support for small businesses**	**Simple dashboards, forecasting, AI assistant, feedback analysis, reports, and alerts**	**Depends on data quality and AI API availability**	**High**

2.3 Need for New System

Based on the analysis of existing systems, there is a clear need for a new system that is designed specifically for small businesses and non-expert users. Existing business intelligence tools are powerful, but they can be difficult to use without technical knowledge. Accounting systems are useful for financial records, but they may not provide full decision support, predictive analytics, customer feedback analysis, and simple AI advice. Enterprise systems are too complex and costly for many small businesses, while spreadsheets are limited and error-prone. [4], [5], [7], [8]

The proposed Intelligent Decision Support System with Predictive Analytics for Small Businesses aims to fill this gap. It will provide a simple web-based platform that combines data entry, file import, dashboards, forecasting, AI assistant, customer feedback analysis, reports, and alerts. This combination makes the system different from ordinary dashboards or accounting tools. [1], [2], [3], [6]

The new system is needed for the following reasons:

1. To simplify business analysis for non-expert users

Many small business owners do not have a strong background in business analysis, accounting, or data science. The proposed system will explain performance using simple charts, summaries, and AI-generated recommendations. [1], [3]

2. To reduce decision-making based on guessing

Without analysis, many business decisions are made based on personal judgment only. The proposed system will help users make decisions based on actual data, trends, and forecasts. [1], [2]

3. To provide affordable decision support

Hiring consultants or using advanced enterprise systems can be expensive. The proposed system will focus on the needs of small businesses and provide useful analysis without unnecessary complexity.

4. To combine multiple features in one platform

The proposed system will combine dashboards, predictive analytics, customer feedback analysis, AI assistant, reports, and alerts. This reduces the need to use several separate tools. [2], [3], [6]

5. To support future planning

Predictive analytics will help small businesses estimate future sales and expenses. This supports better planning for inventory, budgeting, staffing, and marketing. [2]

6. To understand customer feedback

Customer feedback is important for improving products and services. NLP will help the system analyze feedback and identify common problems or positive points. [3]

7. To notify users about important changes

Smart alerts will help users notice problems early, such as increasing expenses, decreasing sales, or negative customer feedback. This allows faster response and better control.

In conclusion, the proposed system is needed because it provides a practical, user-friendly, and intelligent solution for small businesses. It supports users who do not have a business background by transforming raw business data into clear insights, forecasts, reports, and recommendations. This will help small businesses improve decision-making, understand performance, and plan for future growth.

Chapter Three
Project Methodology

3.1 Project Planning

Project planning is an important stage in the development of the Intelligent Decision Support System with Predictive Analytics for Small Businesses. It provides a clear direction for the project and helps organize the work needed to complete the system within the available time. Since this project includes several modules, such as data entry, CSV/Excel file import, dashboards, predictive analytics, customer feedback analysis, AI assistant, automated reports, and smart alerts, proper planning is necessary to make the project achievable.

The main purpose of project planning is to identify the project requirements, estimate the required time, study the feasibility of the system, and divide the work into clear stages. This project is planned over a duration of nine weeks. During this period, the main focus is on proposal preparation, research, analysis, methodology, software design, and presentation preparation. The implementation and testing stages will be completed in Graduation Project 2.

The project planning stage includes the feasibility study and the Gantt chart. The feasibility study explains whether the project can be developed from technical, time, operational, financial, data, and security points of view. The Gantt chart explains the planned tasks and their duration during the nine-week period.

3.1.1 Feasibility Study

A feasibility study is used to determine whether the proposed system can be developed successfully using the available resources, time, tools, and skills. For this project, the feasibility study covers technical feasibility, time feasibility, operational feasibility, financial feasibility, data feasibility, and security feasibility.

Technical Feasibility

The proposed system is technically feasible because the required technologies are available and suitable for developing a modern web-based platform. The frontend of the system will be developed using Next.js and Tailwind CSS. These tools will help create a responsive and user-friendly interface that can be accessed through a web browser.

The backend will be developed using Node.js and Express.js. The backend will handle user requests, business logic, authentication, file processing, forecasting operations, report generation, and communication with the database and AI service. PostgreSQL will be used as the main database because it is reliable and suitable for storing structured business data. Prisma ORM will be used to simplify communication between the application and the database.

The system will also integrate with Google AI API to support the NLP-powered business assistant and customer feedback analysis[12]. Chart.js or Recharts will be used to display dashboards and charts. GitHub will be used for version control, Postman for API testing, and Vercel for deployment.

The main technical challenge is combining several modules into one system. These modules include dashboard visualization, CSV/Excel file parsing, forecasting, NLP analysis, report generation, and alert management. However, by developing the system in stages and focusing on the core features first, the system can be completed successfully.

Financial Feasibility

The project is financially feasible because most of the required tools are free, open-source, or available with free plans. Visual Studio Code, Git, GitHub, PostgreSQL, Prisma ORM, Node.js, Next.js, and Tailwind CSS can be used without major cost. Vercel also provides a free deployment plan that can be used for the prototype stage.

The only possible cost may come from the AI API usage if the free limit is exceeded. However, during the prototype development, the system can be tested using limited API requests and sample data. Therefore, the financial cost of the project is low and suitable for a graduation project.

Time Feasibility

The duration of Graduation Project 1 is nine weeks. This time is enough to complete the proposal, research, background study, methodology, system analysis, software design, and presentation. Since the system includes several modules, the project must be divided into clear tasks. The most important tasks will be completed first, while advanced features can be considered as future work or completed during Graduation Project 2.

The implementation stage will not be fully completed in this chapter. Instead, this phase focuses on planning and designing the system clearly so that the development stage in Graduation Project 2 becomes easier.

Operational Feasibility

The system is operationally feasible because it solves a real problem faced by small business owners and managers. Many small business owners do not have a strong business background and may not understand complex business reports or forecasting models. Therefore, the proposed system is designed to present business information in a simple way using dashboards, summaries, alerts, and AI-generated recommendations.

The system will help users enter or import business data, view business performance, forecast future results, analyze customer feedback, and ask the AI assistant questions using simple language. This makes the system useful for small shops, cafés, restaurants, service providers, startups, and similar small businesses.

Data Feasibility

The system depends on structured business data, such as sales records, expenses, customer feedback, customer numbers, product or service categories, and inventory costs. Since real business data may be difficult to obtain because of privacy and confidentiality, sample datasets can be used during development and testing.

The system will allow users to enter data manually or import CSV/Excel files. To make the analysis accurate, the system must validate uploaded files and ensure that required fields such as date, amount, category, and description are available. Forecasting accuracy will depend on the quality and size of the data. If the data is missing, inconsistent, or too small, the system should inform the user that the forecast may not be accurate.

Security Feasibility

The proposed system will store business-related data, so security must be considered from the design stage. The system should protect user accounts, business records, uploaded files, reports, and chat history. User passwords should be stored securely using hashing. The system should use authentication and authorization to ensure that each user can only access their own business data.

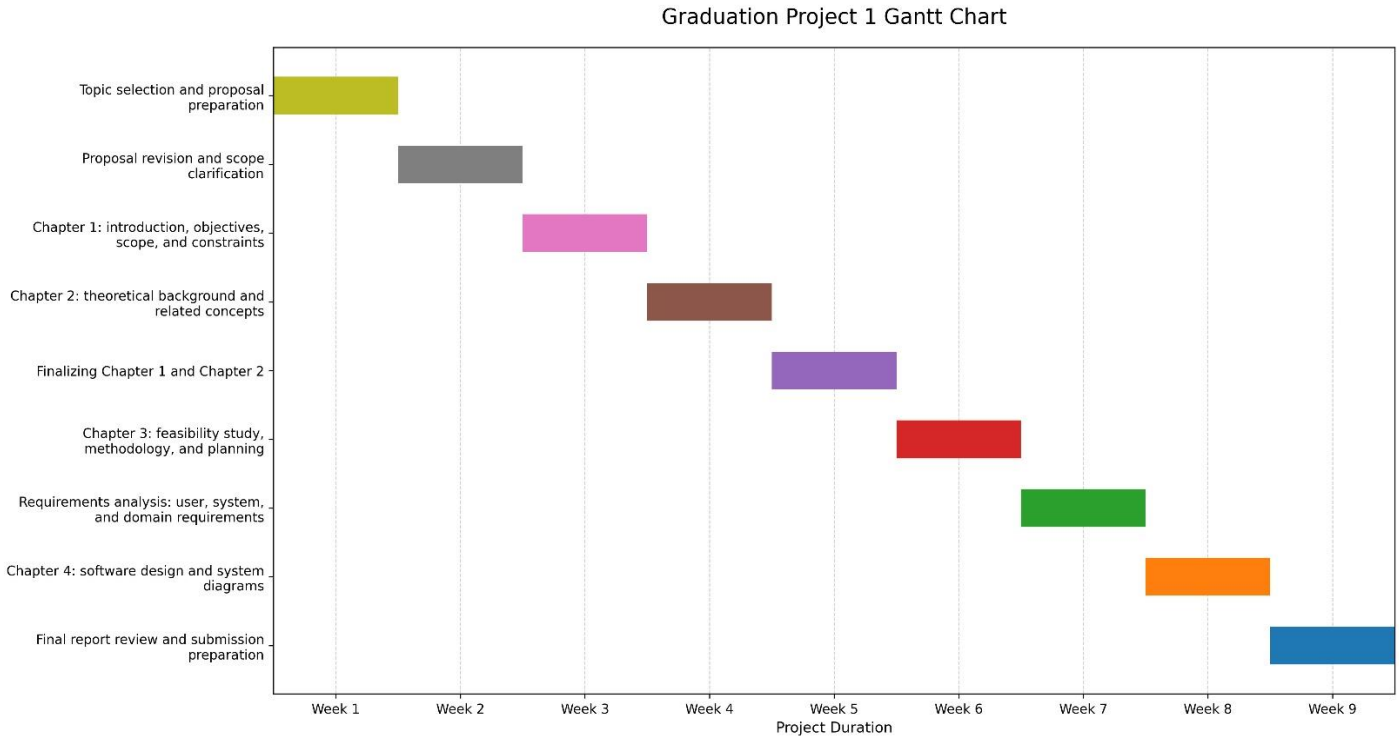
The system should also validate file uploads to reduce the risk of invalid or harmful files. Since the AI assistant will generate advice based on user data, the system should clearly present AI output as decision-support guidance, not as guaranteed financial or legal advice.

3.1.2 Gantt Chart

The project duration is nine weeks. The Gantt chart will show the planned work for each week. The main tasks include proposal preparation, background research, Chapter 1, Chapter 2, Chapter 3, Chapter 4, diagrams, presentation preparation, and final review.

Table 3.1: Project Schedule

Week	Task	Description
Week 1	Proposal preparation	Select the topic and define the title, objectives, description, tools, and technologies.
Week 2	Proposal revision	Edit the proposal based on supervisor feedback and clarify the project scope.
Week 3	Chapter 1	Write the overview, motivation, problem statement, objectives, scope, constraints, and document organization.
Week 4	Chapter 2 background	Research DSS, BI, KPIs, dashboards, predictive analytics, NLP, AI assistant, reports, and alerts.
Week 5	Finalizing Chapter 1 and Chapter 2	Review Chapter 1 and Chapter 2, improve consistency, complete related systems analysis, and finalize references.
Week 6	Chapter 3	Write the feasibility study, software development methodology, project planning, and Gantt chart.
Week 7	Requirements analysis	Define user requirements, functional and non-functional requirements, system requirements, and domain requirements.
Week 8	Chapter 4	Prepare the use case diagram, database design, class diagram, sequence diagram, activity diagram, and architecture diagram.
Week 9	Final report review	Review the report, fix formatting, finalize diagrams, check references, and prepare the final submission.



3.2 Software Development Methodology

A software development methodology is used to organize the process of building a software system. It defines how the project will move from planning and analysis to design, implementation, testing, deployment, and maintenance. Choosing a suitable methodology is important because the proposed system includes different modules and may require changes during development.

For this project, the Agile methodology is suitable because the system can be developed in small stages[9][10]. The project includes many features, such as dashboards, forecasting, AI assistant, reports, alerts, and file import. It is better to build these features gradually instead of trying to complete everything at once. Agile also allows changes to be made when new requirements or improvements are discovered.

3.2.1 Agile Software Development Methodology

Agile is an iterative software development methodology. It focuses on dividing the project into smaller parts and improving the system step by step[9][10]. Each stage produces a working part of the system, and feedback can be used to improve the next stage.

Agile is suitable for this project for the following reasons:

1. The project has several modules that can be developed separately.
2. The requirements may change after supervisor feedback.
3. The AI assistant and forecasting features may need testing and improvement.
4. The system targets non-expert users, so the interface may need changes to make it easier.

5. The project duration is limited, so the most important features should be developed first.

In this project, the main priority is to develop a working prototype that includes the essential features. These features include user authentication, business profile management, data entry, file import, dashboards, forecasting, feedback analysis, AI assistant, reports, and alerts.

3.2.2 Phases of Software Development Methodology

The proposed system will follow the main phases of Agile development. These phases are planning, analysis, design, implementation, testing, deployment, and maintenance[9][10].

Figure 3.2: Agile Software Development Lifecycle Diagram.

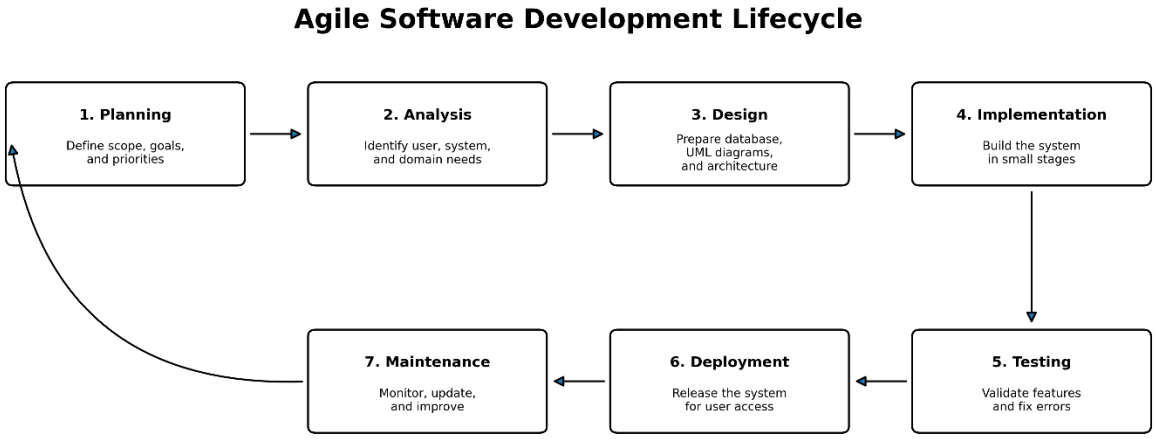


Figure 3.2: Agile software development lifecycle for the proposed system

Planning Phase

In the planning phase, the project idea, objectives, scope, constraints, and required tools are identified. The planning phase also includes the feasibility study and the project schedule. This phase helps ensure that the system can be developed within the available time and resources.

Analysis Phase

In the analysis phase, the system requirements are collected and organized. This includes identifying the target users, user needs, functional requirements, non-functional requirements, system requirements, and domain requirements. The analysis phase is important because it explains what the system should do before starting the design and implementation.

Design Phase

In the design phase, the structure of the system is prepared. This includes designing the database, use case diagram, class diagram, sequence diagram, activity diagram, and system architecture. The design phase helps explain how the system components will interact with each other.

Implementation Phase

In the implementation phase, the actual system will be developed using the selected tools and technologies. The frontend will be developed using Next.js and Tailwind CSS. The backend will use Node.js and Express.js. PostgreSQL and Prisma ORM will be used for database management. Google AI API will be used for the AI assistant and NLP features[12]. This phase will be mainly completed in Graduation Project 2.

Testing Phase

In the testing phase, the system will be tested to ensure that it works correctly. The testing will include checking user login, manual data entry, file import, dashboard calculations, forecasting results, AI assistant responses, report generation, and alert notifications. The purpose of testing is to identify errors and fix them before the final submission.

Deployment Phase

In the deployment phase, the system will be hosted online so users can access it through a web browser. Vercel can be used to deploy the web application. The database and backend services must be configured properly to ensure that the system works smoothly.

Maintenance Phase

In the maintenance phase, the system will be improved after testing and feedback. This may include fixing bugs, improving the interface, improving forecasting accuracy, adding more report types, or improving the AI assistant responses.

3.3 Analysis of the New System

The analysis of the new system defines what the proposed system should do and how it should behave. It includes user requirements, functional requirements, non-functional requirements, system requirements, and domain requirements.

3.3.1 User Requirements

The main users of the system are small business owners, managers, startup founders, and users without a strong business background. These users need a simple system that helps them understand their business performance without requiring advanced knowledge in finance, analytics, or programming.

Table 3.2: User Requirements

Requirement ID	User Requirement
UR1	The user should be able to create an account and log in securely.
UR2	The user should be able to create and manage a business profile.
UR3	The user should be able to enter sales, expenses, customer numbers, inventory costs, and customer feedback manually.
UR4	The user should be able to upload structured CSV or Excel files.
UR5	The user should be able to view dashboards that summarize business performance.
UR6	The user should be able to view important KPIs such as revenue, expenses, profit margin, and sales trends.
UR7	The user should be able to generate sales and expense forecasts.
UR8	The user should be able to analyze customer feedback and identify positive, negative, and neutral comments.
UR9	The user should be able to ask the AI assistant business-related questions in simple language.
UR10	The user should be able to generate periodic business reports.
UR11	The user should receive alerts when important changes or anomalies occur.
UR12	The user should be able to understand system outputs without needing advanced business knowledge.

Functional Requirements

Functional requirements describe the main functions that the system must provide.

Table 3.3: Functional Requirements

Requirement ID	Functional Requirement
FR1	The system should allow users to register, log in, and log out securely.
FR2	The system should allow users to create, update, and delete business profiles.
FR3	The system should allow users to manually enter sales records.
FR4	The system should allow users to manually enter expense records.
FR5	The system should allow users to enter customer feedback.
FR6	The system should allow users to import CSV and Excel files.
FR7	The system should validate uploaded files before storing the data.
FR8	The system should store business data in a structured PostgreSQL database.
FR9	The system should calculate KPIs such as revenue, expenses, profit, and profit margin.
FR10	The system should display KPIs using charts, cards, and dashboards.
FR11	The system should generate sales and expense forecasts using historical data.
FR12	The system should analyze customer feedback using NLP techniques.
FR13	The system should provide an AI business assistant through a chat interface.
FR14	The system should generate business reports for selected time periods.
FR15	The system should create alerts when sales drop, expenses increase, or negative feedback increases.
FR16	The system should allow users to view previous reports, alerts, forecasts, and AI chat history.

Non-Functional Requirements

Non-functional requirements describe the quality attributes of the system.

Table 3.4: Non-Functional Requirements

Requirement ID	Non-Functional Requirement
NFR1	The system interface should be simple and easy to use for non-expert users.
NFR2	The system should be responsive and work on laptops, tablets, and smartphones.
NFR3	The system should load dashboards within a reasonable time.
NFR4	The system should protect user data through authentication and authorization.
NFR5	The system should validate user input to reduce errors.
NFR6	The system should store data reliably in the database.
NFR7	The system should be maintainable and organized using clean code structure.
NFR8	The system should be scalable enough to support more businesses in the future.
NFR9	The system should clearly explain that forecasts and AI recommendations are decision-support information, not guaranteed results.
NFR10	The system should handle errors by showing clear messages to the user.

3.3.2 System Requirements

System requirements describe the hardware, software, and tools needed to develop and run the proposed system.

Hardware Requirements

Table 3.5: Hardware Requirements

Hardware	Requirement
Developer laptop/PC	Used to develop and test the system.
Processor	Intel Core i7.
RAM	16 GB
Storage	500 GB SSD
Internet connection	Required for AI API, deployment, GitHub, and online testing.
Client device	Any laptop, tablet, or smartphone with a modern web browser.

Software Requirements

Table 3.6: Software Requirements

Software/Tool	Purpose
Visual Studio Code	Code editor for development.
JavaScript / TypeScript	Programming languages for frontend and backend.
Next.js	Frontend framework and web application structure.
Tailwind CSS	Styling and responsive interface design.
Node.js	Runtime environment for backend development.
Express.js	Backend API development.
PostgreSQL	Main relational database.
Prisma ORM	Database modeling and communication with PostgreSQL.
Google AI API	AI assistant and NLP support.
Chart.js or Recharts	Data visualization and dashboard charts.
Git and GitHub	Version control and project backup.
Postman	API testing.
Vercel	Web application deployment.

3.3.3 Domain Requirements

Domain requirements describe the special conditions and rules related to the field in which the system will be used. These requirements are important because they make sure that the proposed system works according to the real needs of its target users. In this project, the system is related to the small business decision-support domain. Therefore, the system must support common business data, basic business calculations, customer feedback analysis, forecasting, and simple business recommendations.

The proposed system should be able to handle small business data such as sales records, expenses, products or services, customers, inventory costs, and customer feedback. Sales records should include important information such as date, product or service name, quantity, unit price, and total amount. Expense records should also include date, category, amount, and description so that the system can organize costs and calculate business performance correctly.

Since the system is designed to support business decisions, it should calculate basic Key Performance Indicators (KPIs), such as revenue, expenses, profit, and profit margin. These calculations should be based on simple business formulas, such as subtracting total expenses from total revenue to calculate profit. The system should also use historical business data to generate forecasts, but predictions should be presented as estimated values, not guaranteed results.

The system should support customer feedback analysis using Natural Language Processing. Customer feedback should include the feedback text, date, sentiment result, and detected topic if available. This will help the business owner understand common customer complaints, positive comments, and repeated issues.

Another important domain requirement is that the AI assistant should provide simple business advice based on the available business data. Since the target users may not have strong business knowledge, the system should use simple language in dashboards, reports, alerts, forecasts, and AI responses. Finally, the system should not replace professional accounting, auditing, legal, or business consulting services. It should only provide decision-support guidance to help small business owners understand their situation and make more informed decisions.

Chapter Four
Software Design

4.1 Use Case Diagram

The use case diagram shows how users interact with the proposed system[11]. The main actor is the small business owner or manager. This actor uses the system to manage business data, view dashboards, generate forecasts, analyze feedback, ask the AI assistant, generate reports, and receive alerts.

The system administrator is another actor who may manage user accounts, monitor system performance, and maintain the platform. The external AI service is also included because the system depends on Google AI API to support customer feedback analysis and the AI business assistant.

Figure 4.1: Use Case Diagram for the Intelligent Decision Support System.

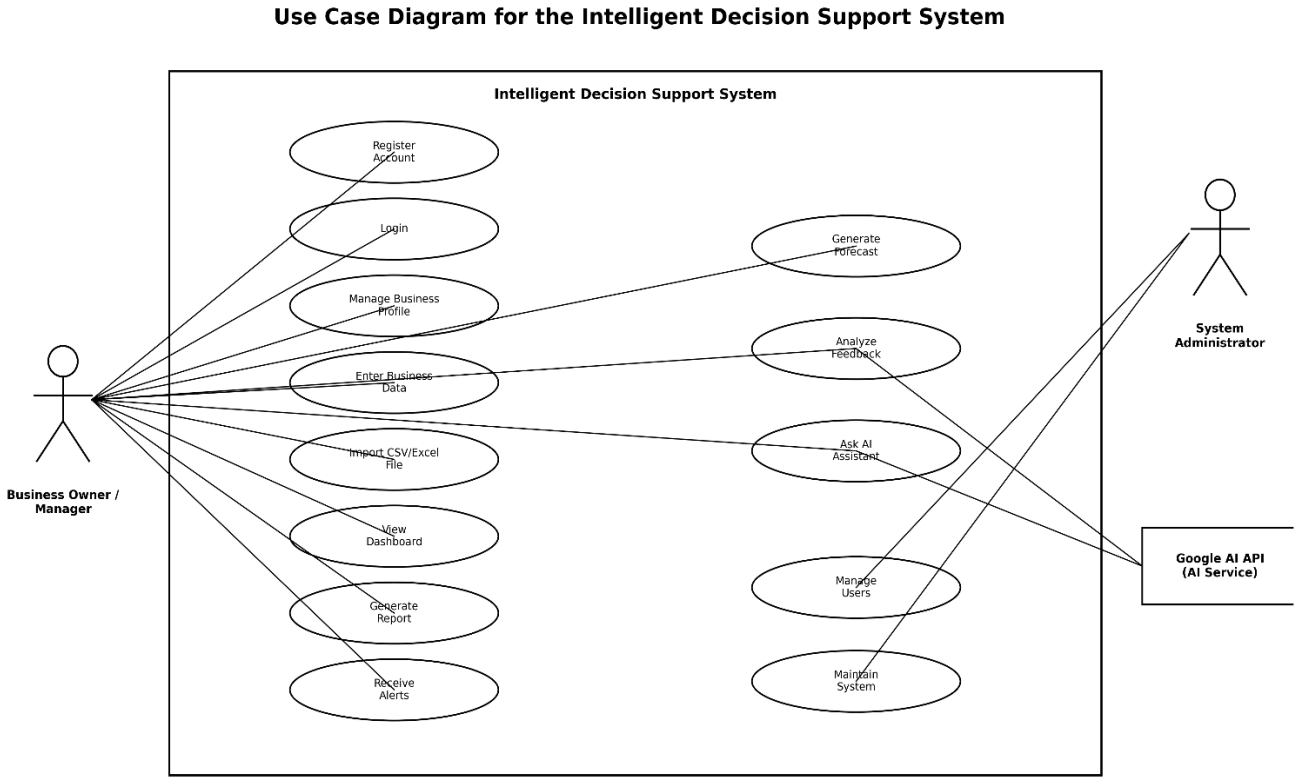


Figure 4.1: Use case diagram for the Intelligent Decision Support System

Use Case Actors Description

The use case diagram includes three main actors: the Business Owner or Manager, the System Administrator, and the AI Service.

The Business Owner or Manager is the main user of the system. This user can register an account, log in, manage a business profile, enter business data, import CSV or Excel files, view dashboards, generate forecasts, analyze customer feedback, ask the AI assistant, generate reports, and receive alerts.

The System Administrator is responsible for managing system users, monitoring the system, and maintaining the platform settings. This actor ensures that the system remains organized, secure, and available for users.

The AI Service represents the external Google AI API used by the system. It supports customer feedback analysis, summarization, and AI assistant responses. This service helps the system provide natural language analysis and simple business advice.

Use Case Explanation

Register Account: The Business Owner or Manager creates a new account to use the system.

Login: The Business Owner or Manager logs in securely to access the system features.

Manage Business Profile: The Business Owner or Manager creates or updates business information, such as business name, business type, and currency.

Enter Business Data: The Business Owner or Manager manually enters business data, such as sales, expenses, inventory costs, customer numbers, and customer feedback.

Import CSV/Excel File: The Business Owner or Manager uploads structured business data from a CSV or Excel file.

View Dashboard: The Business Owner or Manager views business KPIs, charts, summaries, and performance indicators.

Generate Forecast: The Business Owner or Manager generates future sales or expense predictions based on historical business data.

Analyze Feedback: The Business Owner or Manager uses the system to analyze customer feedback. The AI Service helps detect sentiment and common topics from the feedback.

Ask AI Assistant: The Business Owner or Manager asks business-related questions through the chat interface. The AI Service helps generate simple and understandable business advice.

Generate Report: The Business Owner or Manager generates a business performance report for a selected period.

Receive Alerts: The Business Owner or Manager receives notifications about important business changes, such as sales drops, expense increases, or negative feedback patterns.

Manage Users: The System Administrator manages user accounts and system access.

Maintain System: The System Administrator monitors the system and maintains platform settings to ensure that the system works correctly.

4.2 Design of Database

The database design is an important part of the proposed system because the system depends on storing and analyzing structured business data. PostgreSQL will be used as the main database. Prisma ORM will be used to define the database models and manage communication between the application and the database.

The database will store users, businesses, sales records, expenses, customer feedback, imported files, forecast results, reports, alerts, and AI chat messages.

Figure 4.2: Entity Relationship Diagram / Database Design.

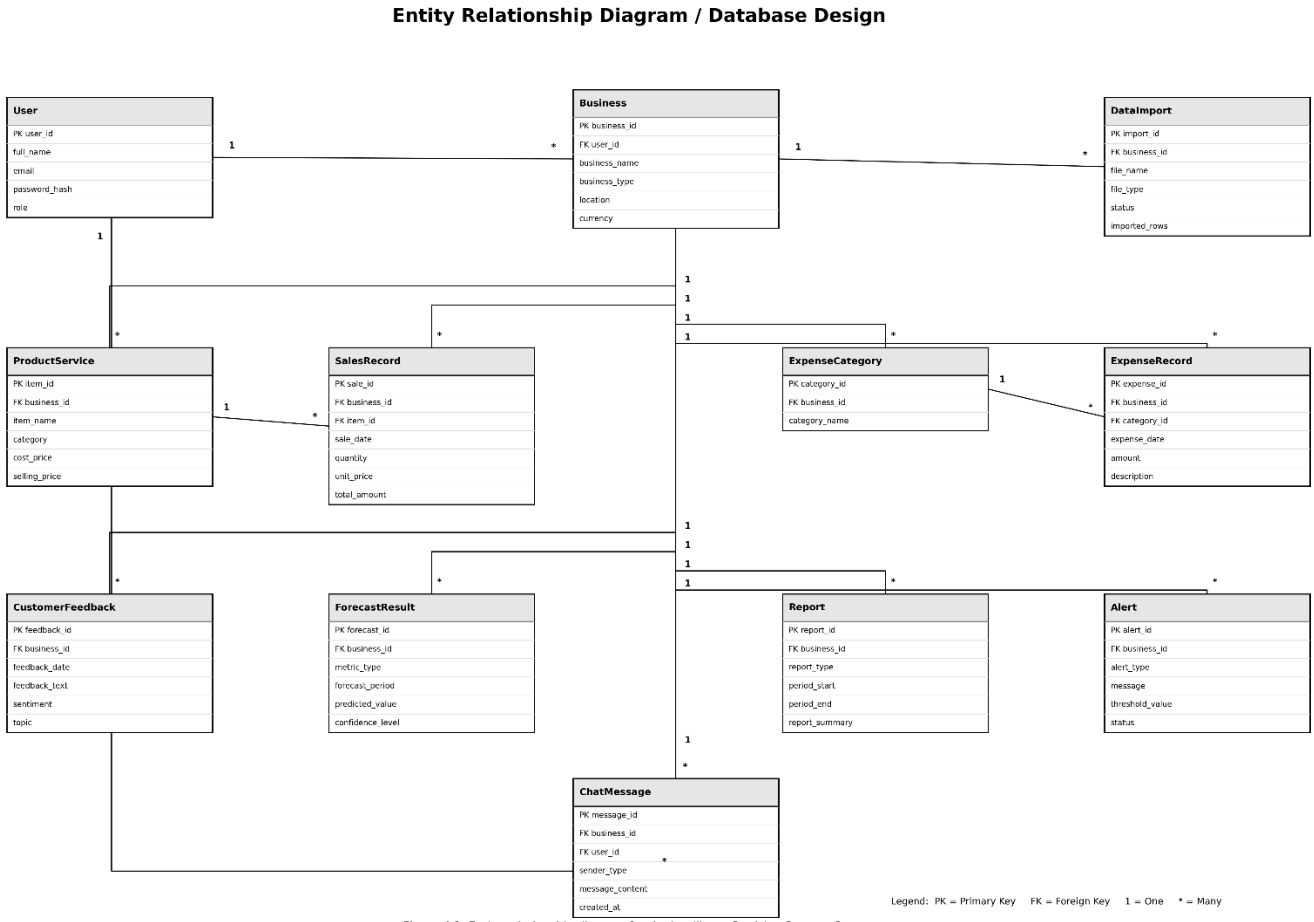


Figure 4.2: Entity relationship diagram for the Intelligent Decision Support System

Main Database Tables

Table 4.3: Database Entities Description

Entity	Description
User	Stores user account details such as name, email, password hash, and role.
Business	Stores information about each business profile.
ProductService	Stores products or services related to the business.
SalesRecord	Stores sales transactions entered manually or imported from files.

ExpenseCategory	Stores expense categories such as rent, salaries, marketing, or supplies.
ExpenseRecord	Stores expense records related to the business.
CustomerFeedback	Stores customer comments, sentiment results, and detected topics.
DataImport	Stores details about uploaded CSV or Excel files.
ForecastResult	Stores generated forecast results for sales or expenses.
Report	Stores generated report summaries and report metadata.
Alert	Stores alerts created by the system based on thresholds or anomalies.
ChatMessage	*Stores chat messages between the user and the AI assistant.*

Database Relationships

The User table has a one-to-many relationship with the Business table because one user may manage more than one business profile. The Business table is connected to most other tables because sales, expenses, feedback, forecasts, reports, alerts, imports, and chat messages belong to a specific business.

The ProductService table has a one-to-many relationship with SalesRecord because one product or service can appear in many sales records. The ExpenseCategory table has a one-to-many relationship with ExpenseRecord because one category can contain many expense records.

The CustomerFeedback table is connected to the Business table because each feedback record belongs to a business. The ForecastResult, Report, Alert, DataImport, and ChatMessage tables are also connected to the Business table to ensure that each business has its own analysis, reports, notifications, imported files, and AI assistant history.

Suggested Database Fields

User
user_id, full_name, email, password_hash, role, created_at, updated_at

Business
business_id, user_id, business_name, business_type, location, currency, created_at, updated_at

ProductService
item_id, business_id, item_name, category, cost_price, selling_price, created_at

SalesRecord
sale_id, business_id, item_id, sale_date, quantity, unit_price, total_amount, created_at

ExpenseCategory
category_id, business_id, category_name, created_at

ExpenseRecord
expense_id, business_id, category_id, expense_date, amount, description, created_at

CustomerFeedback
feedback_id, business_id, feedback_date, feedback_text, sentiment, topic, created_at

DataImport
import_id, business_id, file_name, file_type, status, imported_rows, created_at

ForecastResult
forecast_id, business_id, metric_type, forecast_period, predicted_value, confidence_level, created_at

Report
report_id, business_id, report_type, period_start, period_end, report_summary, created_at

Alert
alert_id, business_id, alert_type, message, threshold_value, status, created_at

ChatMessage
message_id, business_id, user_id, sender_type, message_content, created_at

4.3 Class Diagram

The class diagram shows the main classes of the proposed system and how they are related to each other[11]. The system will include classes for users, businesses, records, feedback, forecasts, reports, alerts, and AI assistant messages. It will also include service classes that handle the main business logic, such as dashboard calculation, forecasting, file parsing, feedback analysis, report generation, and alert management.

Figure 4.3: Class Diagram for the proposed system.

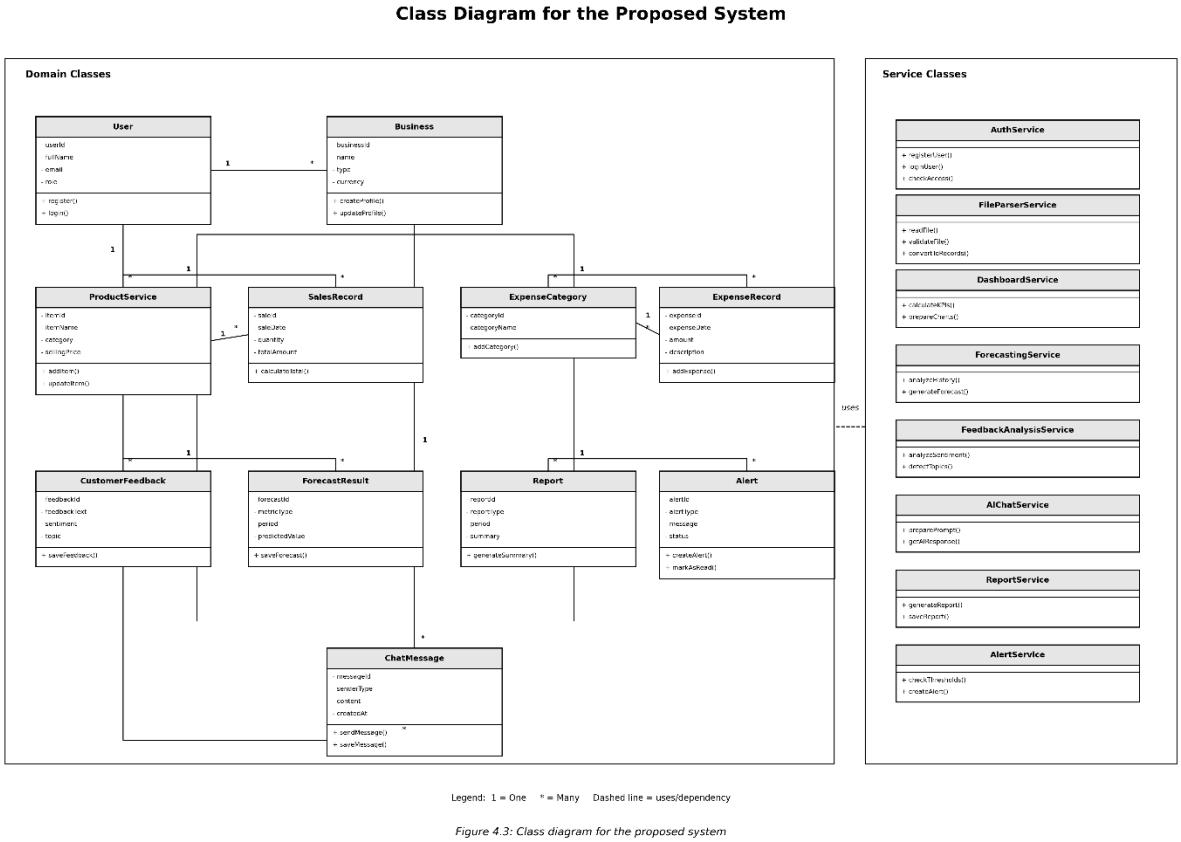


Figure 4.3: Class diagram for the proposed system

The class diagram includes two main groups: domain classes and service classes. The domain classes represent the main data objects used in the system, such as User, Business, SalesRecord, ExpenseRecord, CustomerFeedback, ForecastResult, Report, Alert, and ChatMessage. These classes store the important information that the system needs to manage business data, generate reports, create alerts, and support AI assistant conversations.

The service classes represent the main system operations. AuthService handles user registration, login, and access control. FileParserService manages CSV and Excel file reading, validation, and conversion into database records. DashboardService calculates KPIs and prepares chart data. ForecastingService generates predictions from historical sales and expense records. FeedbackAnalysisService analyzes customer feedback using NLP techniques. AIChatService handles user questions and AI assistant responses. ReportService generates business reports, while AlertService checks important changes and creates alerts.

The service classes help separate the system logic from the database models. This makes the system easier to maintain and improve. For example, if the forecasting method is changed in the future, the change can be made inside the ForecastingService without affecting the whole system.

4.4 Sequence Diagram

The sequence diagram shows the order of interaction between the user, frontend, backend, database, and AI service[11]. Since the AI business assistant is one of the main features of the system, the sequence diagram should focus on the process of asking the AI assistant for business advice.

Figure 4.4: Sequence Diagram for AI Business Assistant Request.

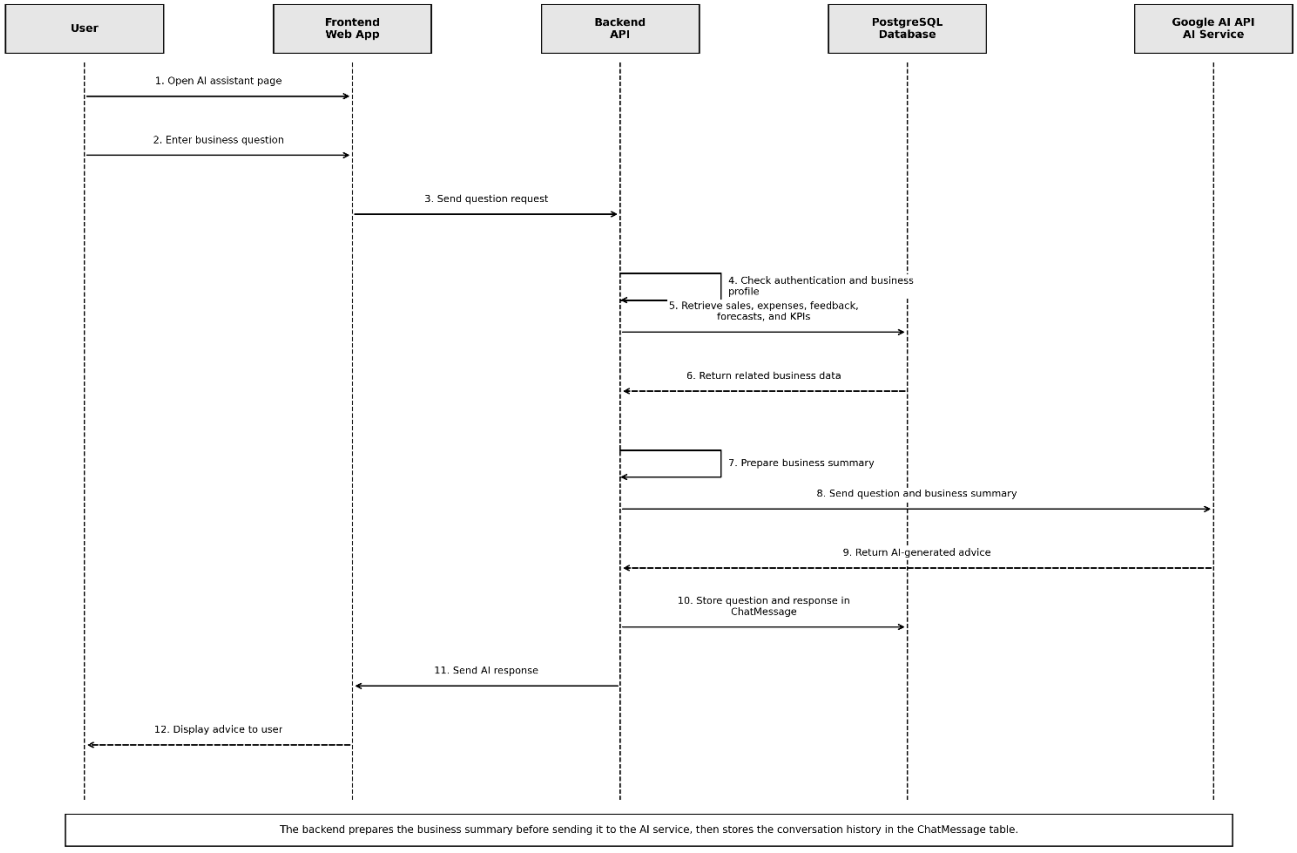


Figure 4.4: Sequence diagram for AI Business Assistant request

Sequence Diagram Description

1. The user logs in and opens the AI assistant page.
2. The user writes a business question, such as “Why did my profit decrease this month?”
3. The frontend sends the question to the backend API.
4. The backend checks the user authentication and business profile.
5. The backend retrieves related business data from the PostgreSQL database.
6. The backend prepares a summary of relevant sales, expenses, customer feedback, forecasts, and KPIs.
7. The backend sends the user question and business summary to Google AI API[12].
8. The AI service generates a simple response.
9. The backend stores the user question and AI response in the ChatMessage table.

10. The frontend displays the AI assistant response to the user.

This sequence ensures that the AI assistant gives advice based on the user's actual business data. It also stores the conversation history so the user can review previous advice later.

4.5 Activity Diagram

The activity diagram shows the workflow of the proposed system from the user's point of view[11]. It explains how the user starts by logging in, enters or imports data, views analysis, generates forecasts, receives alerts, and asks the AI assistant for advice.

Figure 4.5: Activity Diagram for the proposed system.

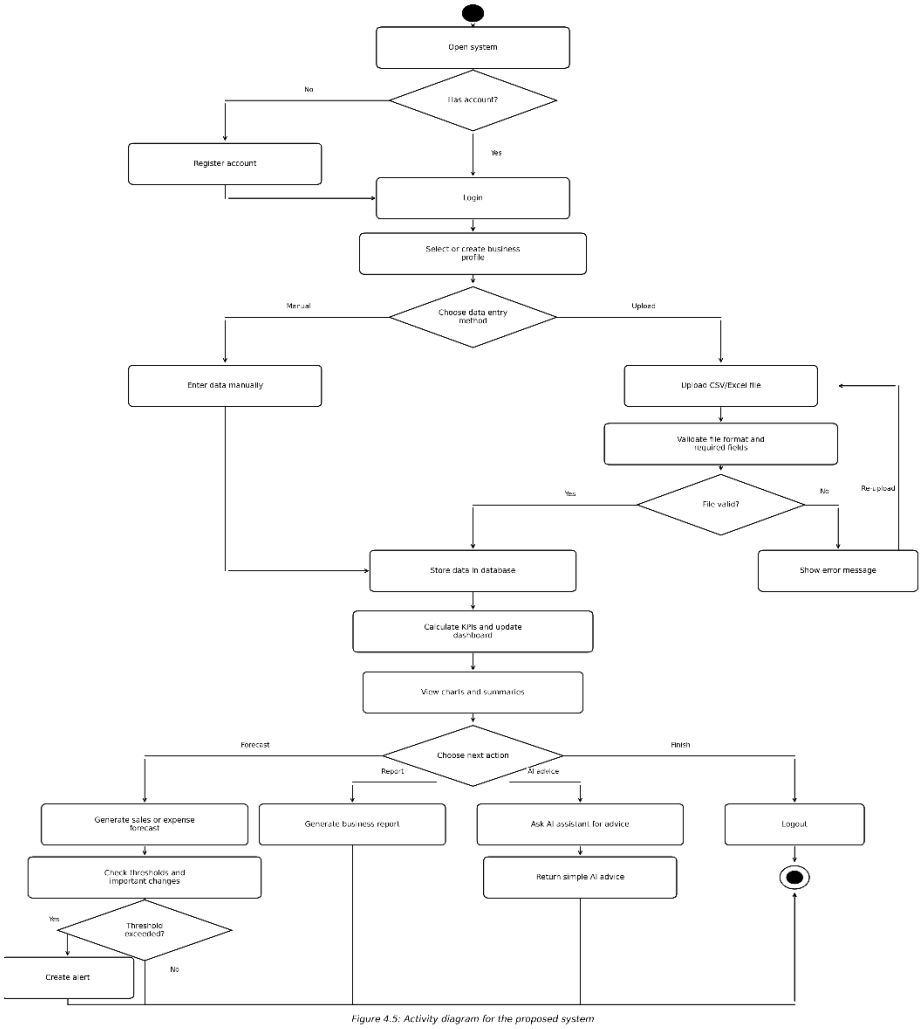


Figure 4.5: Activity diagram for the proposed system

Activity Diagram Description

The activity starts when the user opens the system. If the user does not have an account, the user can register. If the user already has an account, the user logs in. After login, the user selects or creates a business profile.

The user can then choose to enter data manually or upload a CSV/Excel file. If the user uploads a file, the system validates the file format and checks the required fields. If the file is invalid, the system shows an error message. If the file is valid, the data is stored in the database.

After the data is stored, the system calculates KPIs and updates the dashboard. The user can view charts and summaries. The user can also generate forecasts for sales or expenses. The system checks the forecast and KPI results to detect important changes. If a threshold is exceeded, the system creates an alert.

The user can also generate a business report or ask the AI assistant a question. The AI assistant uses available business data and returns simple advice. The process ends when the user logs out.

Chapter Five
Software Implementation

This chapter presents Basira, the implemented Intelligent Decision Support System with Predictive Analytics for Small Businesses. It describes the software and hardware, client-server communication, database mapping and application interfaces. It also explains the refinements made to the design in Chapters 3 and 4.

5.1 Software and Hardware

5.1.1 Software

Table 5.1 lists the installed technologies used in Basira's full-stack TypeScript application and their responsibilities. Versions refer to the project dependencies.

Table 5.1: Software used in the implementation.

Software / version	Implementation purpose
Next.js 16.3.8 and React 19.3.0	Pages, layouts, interactive components and server Route Handlers in one application [13].
TypeScript 6.0.3 and Node.js 24.x	Typed client/server code and the server execution environment.
Tailwind CSS 4.3.3 and Radix UI	Responsive styling, reusable controls and accessible dialogs.
PostgreSQL 18 and pgAdmin 4	Relational storage and local database administration.
Prisma ORM 7.10.0 and pg 8.23.0	Generated database client, schema migrations and PostgreSQL driver adapter [14].
NextAuth.js 4.24.15	Email/password and Google sign-in integration, with JWT sessions [15].
Recharts 3.10.1	Interactive financial, customer and forecast charts.
Papa Parse 5.7.0 and ExcelJS 4.4.0	CSV and XLSX import parsing.
Zod 4.6.5	Runtime validation of requests, imported values and AI output.
NVIDIA API with Google Gemma	Server-side access to google/gemma-4-31b-it for language tasks [16].
React PDF renderer 4.9.0	Local generation of downloadable business-report PDFs [17].

Development uses npm, Git, ESLint, Prettier, Vitest and Playwright for dependency management, version control, code quality and automated verification. Detailed test scenarios and results belong in Chapter 6. GitHub Actions and Vercel configuration are prepared; public deployment and a remote CI run are not claimed.

5.1.2 Hardware

The local prototype runs the browser, Next.js server and PostgreSQL service on the same development laptop. The observed machine configuration is shown in Table 5.2. These are the resources used for the local implementation, not experimentally established minimum requirements for all deployments.

Table 5.2: Hardware and execution environment.

Component	Observed or required environment
Development computer	Intel Core i7-12700H, 14 cores and 20 logical processors; Windows 11 Home, 64-bit.
Memory and storage	Approximately 16 GB RAM (15.7 GiB usable) and a nominal 1 TB SSD (about 954 GiB reported).
Client side	Laptop, desktop, tablet or smartphone with a modern browser, JavaScript enabled and access to the application server.
Local server side	The development laptop hosts Node.js and PostgreSQL 18. The application uses localhost:3000; PostgreSQL uses the configured local port 5433.
External services	An internet connection is required for Google OAuth and NVIDIA inference. No local model-serving GPU is required by Basira.
Future hosted server	A Node-compatible host and PostgreSQL service are required. Production memory, concurrency and capacity must be measured after hosting is configured.

The browser displays forms, tables and charts and sends requests to the application. It does not connect directly to PostgreSQL. Server-side code performs authorization, validates requests, calculates authoritative figures and accesses stored records. NVIDIA operates the external model infrastructure; its hardware is outside the local project environment.

5.2 Implementation

The implementation follows the project modules in stages, using shared services for ownership checks, dates, monetary arithmetic and database access. The screenshots in this chapter show the actual interface with synthetic demonstration data. Captions identify the archived automated-test examples; their simulated AI replies demonstrate interface behavior and do not establish live model accuracy.

5.2.1 Client and Server Architecture

Next.js supplies both the web interface and the server API. Interactive React components manage forms, filters and charts, while server-rendered pages load authorized initial content. Versioned endpoints under /api/v1 handle business operations. Route Handlers use HTTP request and response objects [13]; domain services keep financial and business rules separate from page rendering.

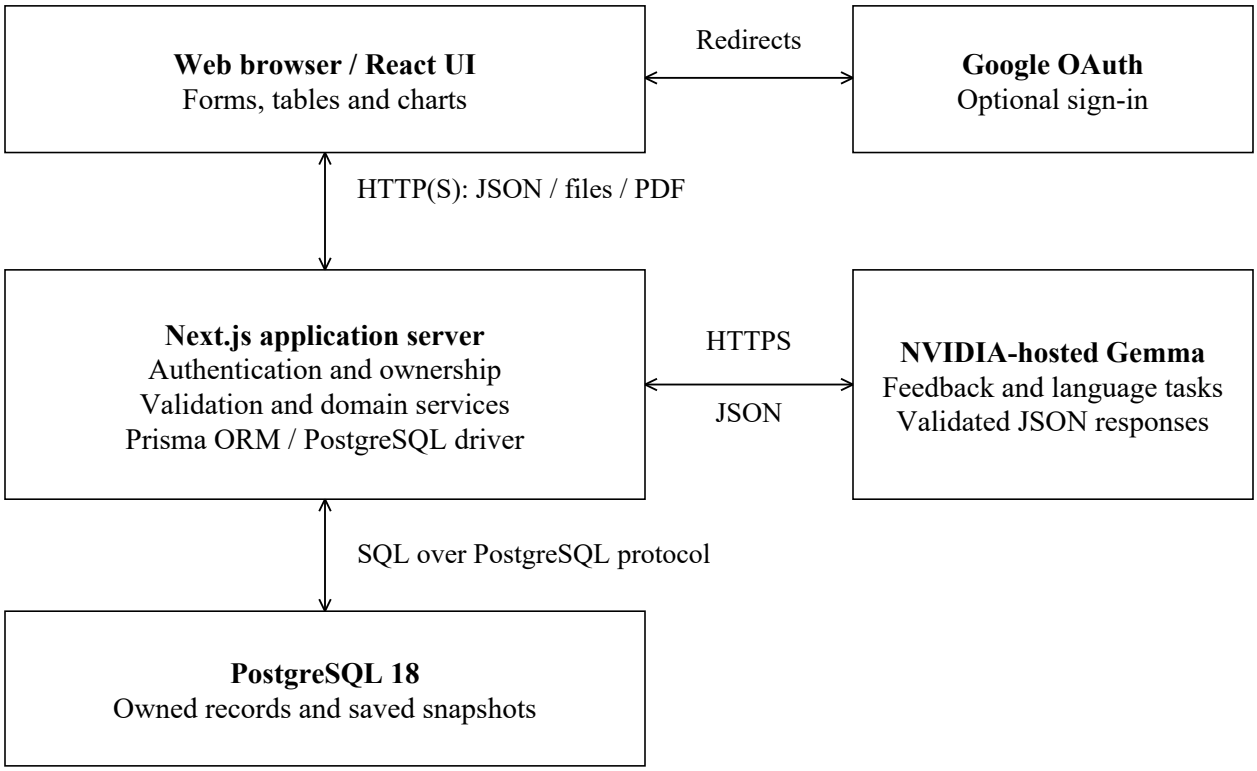


Figure 5.1: Implemented client-server communication and external services.

For a normal record operation, the browser sends a JSON request to the application origin. The server reloads the signed-in user, checks active status and business ownership, validates the input, and executes the relevant service. Prisma and the PostgreSQL driver read or write the database. The response contains only the fields needed by the interface, with monetary values serialized as decimal strings and business dates as YYYY-MM-DD.

CSV/XLSX uploads use multipart form data, while report downloads return application/pdf. Local development uses HTTP on the loopback interface; a hosted deployment must use HTTPS. Google sign-in redirects the browser through the OAuth provider and returns to the application callback. NVIDIA requests are separate server-to-server HTTPS calls with a server-held API key. The model has no direct connection to the database and cannot execute SQL or change records.

This arrangement keeps the client lightweight and avoids a separate Express server. The browser receives useful results without receiving database credentials, OAuth client secrets or the NVIDIA API key.

5.2.2 Object Relational Mapping

Prisma maps the application models in prisma/schema.prisma to PostgreSQL tables and provides a generated TypeScript client. The project uses the Prisma 7 PostgreSQL driver-adapter configuration [14]. Model relations express ownership and links between records; migrations create the corresponding tables, indexes and constraints. The ORM operates between the application server and the database. JSON serialization is a separate step between the server and the browser.

Table 5.3: Main implemented model relationships.

Models	Mapping and responsibility
User, Account and Business	A user owns multiple businesses. OAuth accounts refer to a user. Business ownership scopes the operational data.
ProductService and SalesRecord	A product/service belongs to a business and may be referenced by multiple sales. Each sale retains historical name, price and cost information.
ExpenseCategory and ExpenseRecord	An expense refers to a category in the same business and stores its recorded expense classification.
CustomerMetric and CustomerFeedback	Daily aggregate visits and dated customer comments belong to one business. Each business/date has at most one visit-total row.
DataImport	Stores confirmed import metadata and links imported records to their source operation; raw uploaded files are not retained.
ForecastRun and ForecastPoint	A run stores its model, history and evaluation snapshot; related points store explicitly dated future amounts.
ChatConversation and ChatMessage	Conversations belong to an owner and business; ordered messages retain replies, status and the context used for an attempt.
AlertRule, Alert and Report	Rules configure checks, alerts preserve trigger evidence, and reports store immutable business-summary snapshots.

Financial columns use PostgreSQL numeric types. Prices and amounts use Decimal(19,4), quantities use Decimal(18,3), and an unknown unit cost remains null. Calendar dates use date columns, while creation and processing times use timezone-aware timestamps. The client receives formatted date labels and decimal strings, so browser floating-point arithmetic does not determine saved totals.

For example, a sale request identifies an owned business and an item. The service checks that the item belongs to that business, calculates quantity multiplied by unit price, and saves the total and historical cost in a transaction. Composite foreign keys reject cross-business references. Business-row locks coordinate financial writes and settings changes; record versions reject stale edits.

5.2.3 Refinements to the GP1 Design

Chapters 3 and 4 describe the proposed design. During GP2, several details were refined to make that design executable and to preserve the meaning of business records. Table 5.4 summarizes the principal differences. The project maintains the implementation decision log in docs/GP1_TO_IMPLEMENTATION_CHANGES.md.

Table 5.4: Principal refinements from design to implementation.

GP1 design	Implemented refinement and reason
Next.js with Express backend	One Next.js application with Route Handlers and server services reduces deployment and integration complexity.
Google AI API	NVIDIA-hosted google/gemma-4-31b-it is used through a server-only adapter. Google OAuth remains a separate authentication service.
Basic email/password access	Password sign-in and optional Google OAuth use the same user/ownership checks. Matching email addresses do not automatically link accounts.
Simplified profit formula	Estimated profit subtracts known historical COGS and Operating expenses. Inventory purchases and Other spending are shown separately.
Unspecified forecast method and one result entity	Local Holt forecasting uses recorded-history requirements and a saved run with dated points, holdout errors and a naive baseline.
Customer counts without a separate entity	CustomerMetric records one daily visit total, including repeat visits, without collecting customer identities.
Direct chat messages and alerts	ChatConversation groups messages; AlertRule separates configurable thresholds from generated alert evidence.
Editable business currency	Currency and timezone are permanently locked once financial data is written, preventing reinterpretation of existing amounts or dates.
General automated reporting	Owners generate weekly, monthly or custom snapshots on demand. Calculations and PDF creation are automated; scheduled delivery is not implemented.

These refinements retain the original decision-support purpose. Basira remains a small-business prototype rather than a full accounting or stock-control system. Refunds, negative adjustments, multi-item receipts, currency conversion and inventory quantity movements are outside the implemented scope.

5.2.4 User Authentication

The authentication interface provides account registration, email/password sign-in, Google sign-in when configured, and sign-out. Figure 5.2 shows the sign-in screen. New registrations create normal active users; administrator rights are never selected during registration. Authentication is integrated through NextAuth.js [15].

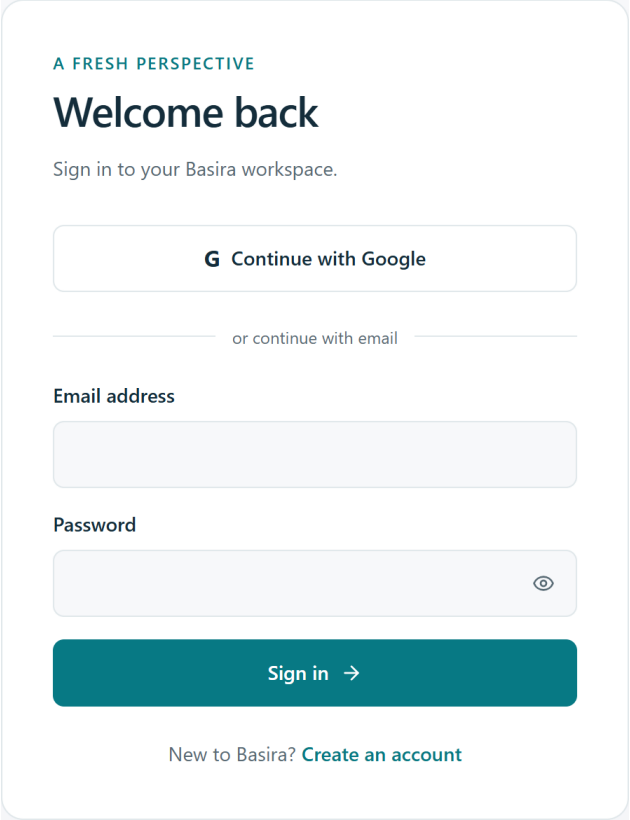


Figure 5.2: Basira sign-in interface with password and Google options.

Passwords are salted and hashed using asynchronous scrypt before storage. The application uses an eight-hour JWT session, with current account status and role reloaded from the database for protected operations. A suspended account cannot make subsequent protected requests using an old session. Shared attempt limits and generic sign-in errors reduce information disclosure.

Google-only accounts can have no password hash. The Google flow requires a verified provider email, valid OAuth credentials and an authorized callback. Equal email addresses alone do not link a new Google identity to an existing password account. The interface confirms that both methods are integrated; the visible Google button does not by itself prove a completed external OAuth exchange.

5.2.5 Business Profiles

After signing in, the owner creates or selects a business. Each profile stores its name, type, optional location, currency and IANA timezone. The active-business selector changes the workspace context, but the selection cookie does not grant access: server services still verify ownership for every operation. Figure 5.3 shows the two synthetic demonstration businesses.

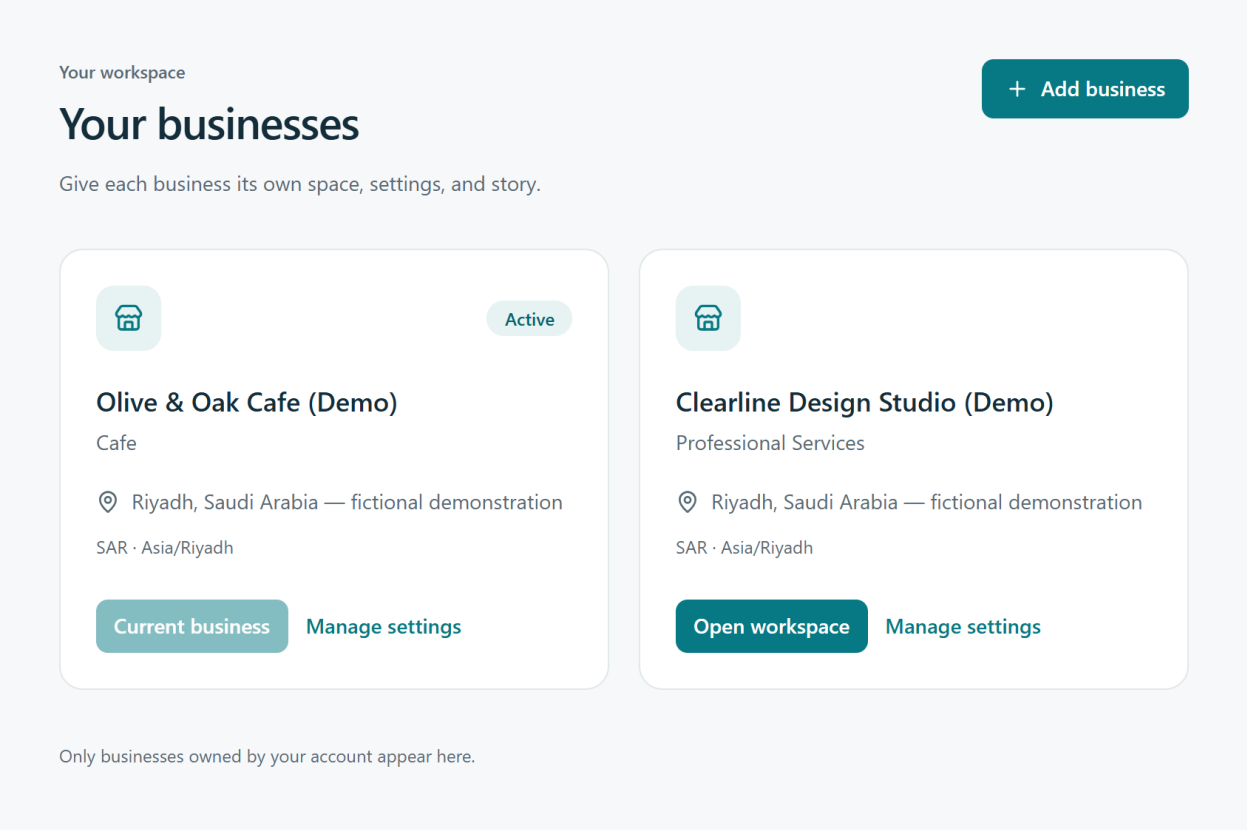


Figure 5.3: Separate business profiles in the demonstration account.

Profile edits require the current version. Deletion requires the current version and the exact business name, and removes the owned record graph. Currency and timezone become locked when financial records, priced catalogue entries or saved monetary snapshots are written. Deleting those records does not clear the lock. No automatic exchange-rate conversion is performed.

5.2.6 Product and Service Catalogue

The catalogue stores item name, product/service type, category, current selling price and optional unit cost. Search, filters and pagination help the owner browse items. An item can be archived when it is no longer offered. Historical sales retain their recorded item name and unit cost, so changing a catalogue price does not rewrite past performance. A blank cost means unknown, while an explicit zero is a known zero cost.

5.2.7 Sales and Expense Records

A sales entry represents one complete transaction for one product or service. The form records the date, item, quantity, unit price, historical unit cost and optional note. The server calculates the total using decimal arithmetic and rounds once to the currency precision. For example, three items sold at SAR 20 each produce a recorded sale total of SAR 60. The number of sale entries is therefore the transaction count used by the average-sale-value KPI.

An expense records its date, category, total amount spent and description. Amount means money, not quantity: buying ten pens at SAR 2 each is an expense of SAR 20. The category supplies one of three classifications: Operating, Inventory Purchase or Other. That treatment is preserved on the expense record. Figure 5.4 shows the entry dialog.

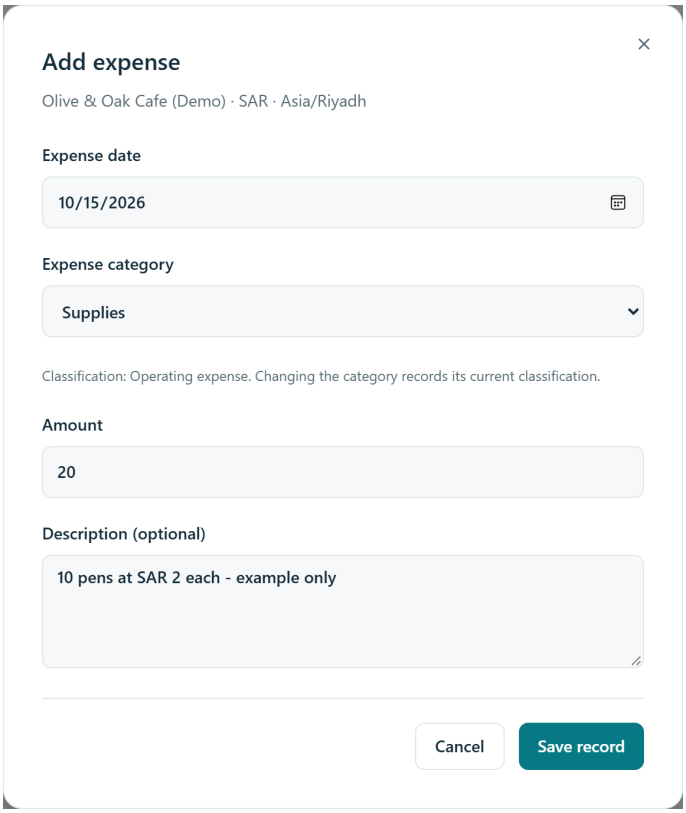


Figure 5.4: Expense-entry form with amount, category and date.

Forms validate calendar dates, required fields, precision and nonnegative sale values; expenses must be positive. Refunds and negative adjustments are deferred until their accounting rules are defined. Edits and deletions require current record versions. Product/category references must belong to the selected business, and referenced categories cannot be deleted until their expenses are reassigned.

5.2.8 Customer Visit Metrics

Customer visits are recorded as one nonnegative whole-number total for each business date. The total includes repeat visits and does not represent unique customers. A unique business/date constraint prevents two entries from silently doubling a day. Correcting an existing total requires an explicit edit rather than importing another row for the same date.

The dashboard sums recorded visits and reports how many dates have counts. An explicit zero is a recorded observation; a missing date remains unknown. This distinction supports a clearer comparison between periods and avoids presenting an incomplete recording history as a measured decline in customer activity. No customer names, telephone numbers or individual customer profiles are required.

5.2.9 CSV and Excel Imports

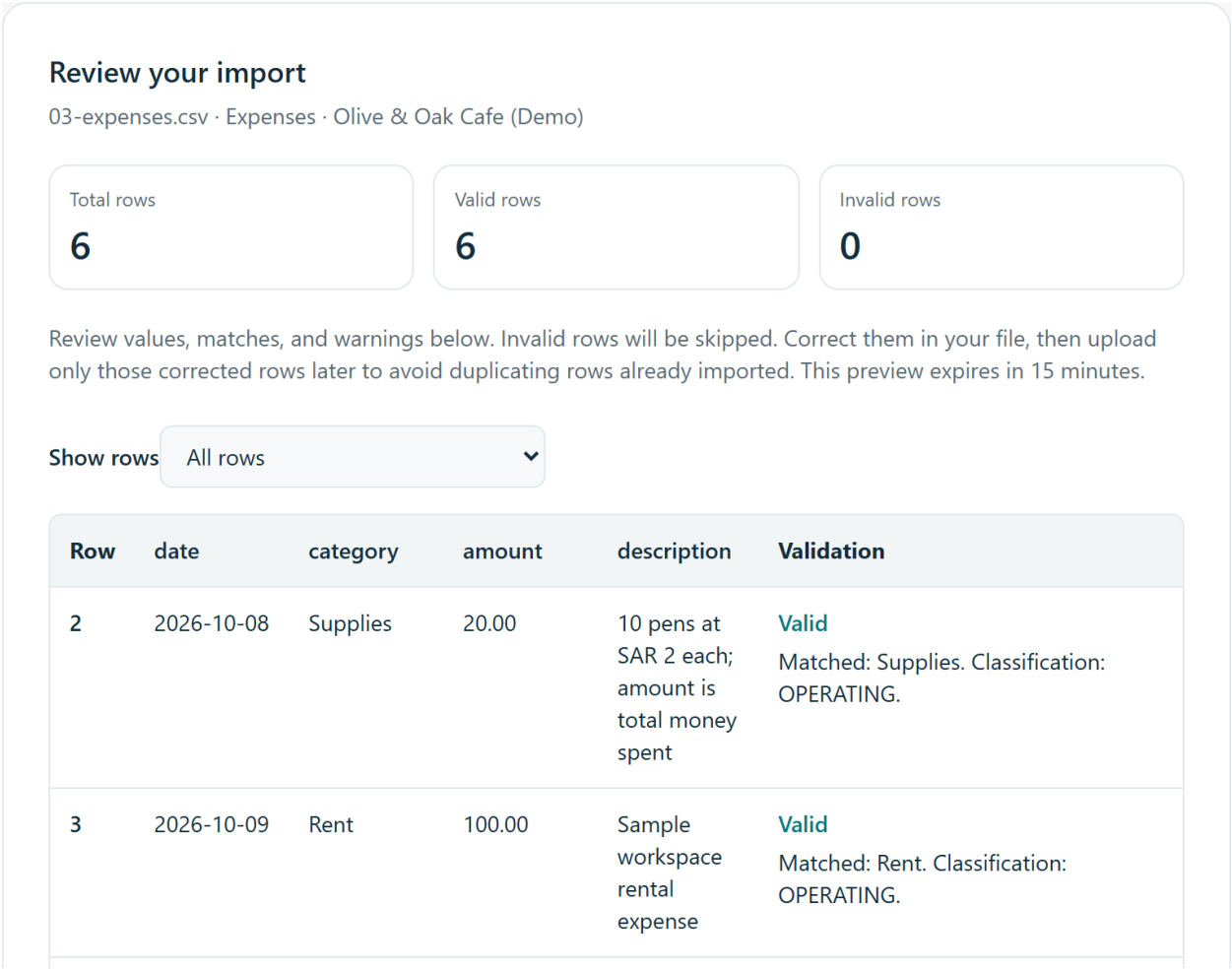
The import workspace provides CSV and XLSX templates for products/services, sales, expenses, customer visits and feedback. The owner chooses the dataset type and uploads one file. The server parses the file and returns a row-by-row preview before any operational records are saved. This implements the file-import and validation requirements from Chapter 3.

Table 5.5: Supported import data and principal validation rules.

Data type	Required information and matching rule
Products / services	Item name, type and selling price. Existing or repeated item names are invalid; imports do not overwrite the catalogue.
Sales	Date, item name, quantity and unit price. The item must uniquely match an active catalogue entry. Historical unit cost may be blank.
Expenses	Date, existing category and amount. The category must already have a defined expense classification.
Customer visits	Date and whole-number customer_count. Existing daily totals require a normal edit.
Feedback	Date and feedback_text. Imported comments remain pending until the owner requests analysis.

Files are limited to 2 MiB and 1,000 data rows. CSV uses UTF-8. XLSX must contain one plain worksheet with values rather than formulas. Extra worksheets, unsafe archive structures, merged cells and unsupported workbook features are rejected. Matching ignores surrounding spaces and letter case, but ambiguous matches remain invalid.

Figure 5.5 shows an expense-file preview. The server reports total, valid and invalid row counts, then displays resolved categories and warnings. Inventory Purchase and Other rows can be valid while warning that their amounts are excluded from estimated profit. An invalid row carries a correction message instead of being silently repaired.



Review your import

03-expenses.csv · Expenses · Olive & Oak Cafe (Demo)

Total rows	Valid rows	Invalid rows
6	6	0

Review values, matches, and warnings below. Invalid rows will be skipped. Correct them in your file, then upload only those corrected rows later to avoid duplicating rows already imported. This preview expires in 15 minutes.

Show rows: All rows

Row	date	category	amount	description	Validation
2	2026-10-08	Supplies	20.00	10 pens at SAR 2 each; amount is total money spent	Valid Matched: Supplies. Classification: OPERATING.
3	2026-10-09	Rent	100.00	Sample workspace rental expense	Valid Matched: Rent. Classification: OPERATING.

Figure 5.5: Expense import preview excerpt; six rows are valid and none has been confirmed.

Confirmation requires an explicit review acknowledgment. A signed preview expires after 15 minutes. Before saving, the server locks the owned business and revalidates the reviewed state, preventing an outdated preview from applying against changed catalogue or category data. The valid subset and its DataImport history are saved in one transaction; an insertion failure rolls back the operation.

The import history records file/type metadata, counts and completion status. An exact-file hash prevents repeat confirmation of the same file within the same business and dataset type. This is not universal duplicate detection: a changed file can still contain transactions already imported. Owners should therefore upload only corrected skipped rows after a partial import. Raw uploaded files and unconfirmed previews are not retained.

5.2.10 Dashboard and Key Performance Indicators

The Overview page summarizes the selected business using period filters for the last 7, 30 or 90 days and a custom interval. It compares each interval with the immediately preceding interval of equal length. Calculations use business-local date labels and authorized database aggregates. Figure 5.6 shows the displayed indicators for the synthetic cafe.

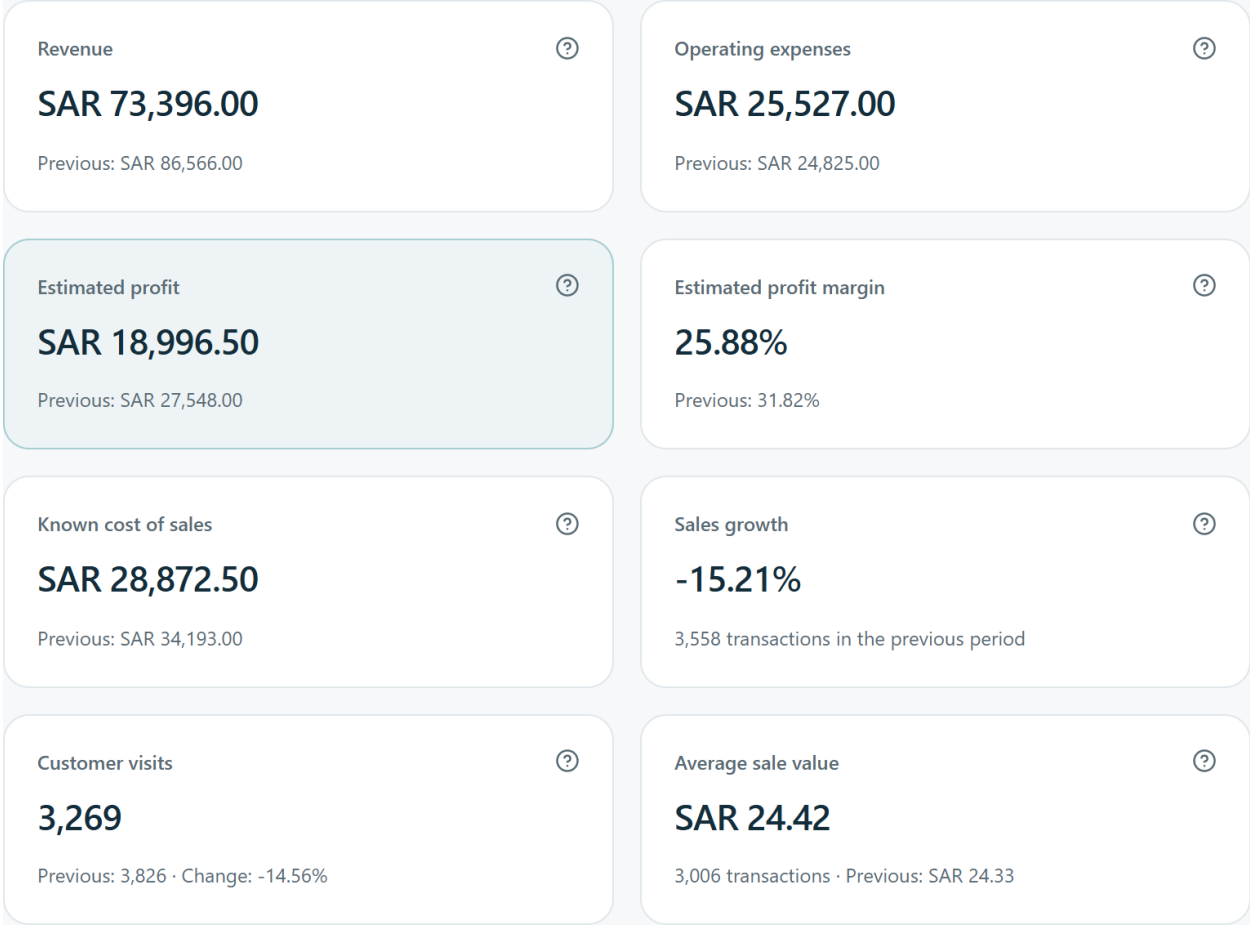


Figure 5.6: Dashboard KPI cards for the synthetic cafe, 18 July to 15 October 2026.

Revenue = sum of recorded sale totals

Known COGS = sum of quantity × historical unit cost for sales with known costs

Estimated profit = Revenue − Known COGS − Operating expenses

Estimated profit margin = Estimated profit ÷ Revenue × 100

Inventory purchases are excluded from the Operating deduction because the cost of sold goods is already represented by historical unit costs. Other spending is also disclosed separately instead of receiving an assumed accounting treatment. Margin is unavailable when revenue is zero. Average sale value divides revenue by the number of recorded single-product/service transactions; growth is unavailable when the comparison baseline is zero.

Revenue, expenses and estimated profit are plotted by date, with separate customer trends, top products and category breakdowns. Figure 5.7 shows the financial trend chart. The chart summarizes recorded amounts; a zero on a date with no financial rows is not proof that the business had no activity.

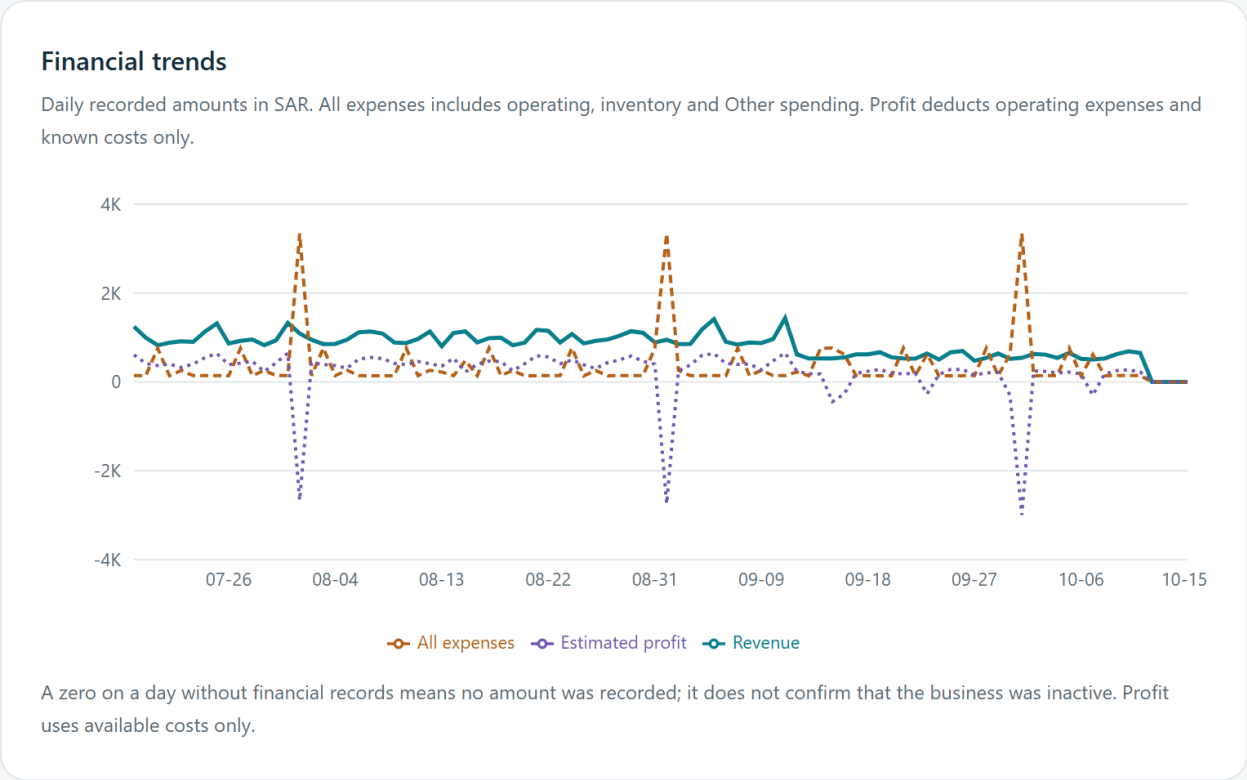


Figure 5.7: Recorded revenue, all expenses and estimated profit over time.

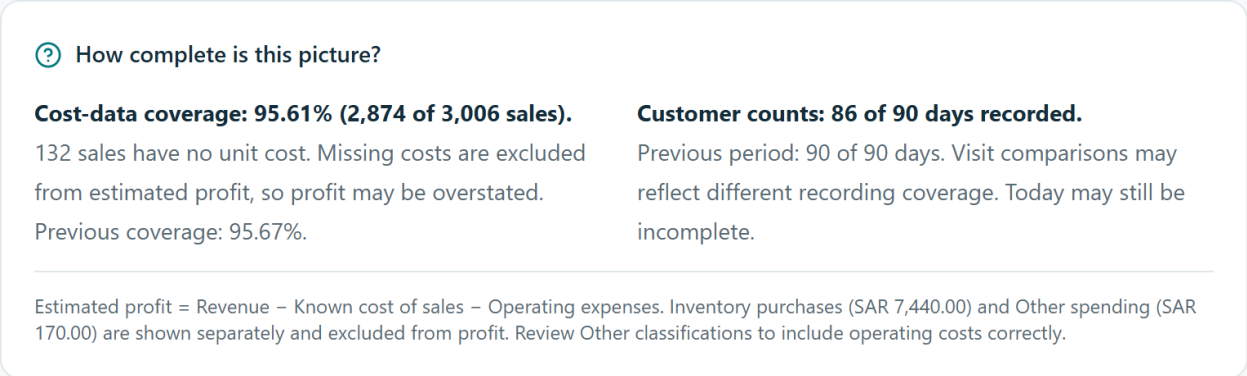


Figure 5.8: Cost and customer-record coverage displayed beside the dashboard.

Cost-data coverage is the number of sales with a known historical unit cost divided by the number of sales in the period. A known zero cost counts as covered; a missing cost does not. In this example, 2,874 of 3,006 sales have costs, giving 95.61% coverage. The 132 remaining sales can cause estimated profit to be overstated. Customer counts cover 86 of the 90 selected dates. These disclosures make incomplete data visible rather than presenting all summary values as equally reliable.

5.2.11 Predictive Analytics

The forecasting module predicts revenue or all recorded expenses using a local implementation of Holt's linear trend method. It does not call the language model. Revenue uses sale totals, while expense forecasts include Operating, Inventory Purchase and Other amounts and therefore describe spending rather than profit. The owner reviews the historical coverage before generation.

Table 5.6: Forecast horizons and implemented history requirements.

Horizon	Recorded dates required	Chronological holdout	Displayed result
Next 7 days	14	7 days	Seven daily amounts
Next 30 days	60	30 days	Thirty daily amounts
Next 3 months	180	90 days	Three full calendar-month totals

History includes at most 366 completed business-local dates and ends yesterday. The minimum counts actual dates containing records. Missing dates cannot qualify a sparse dataset by themselves. The owner must confirm that listed gaps really mean zero activity before they can be used as zero observations; otherwise the missing records should be entered. Figure 5.9 illustrates the history-review interface.

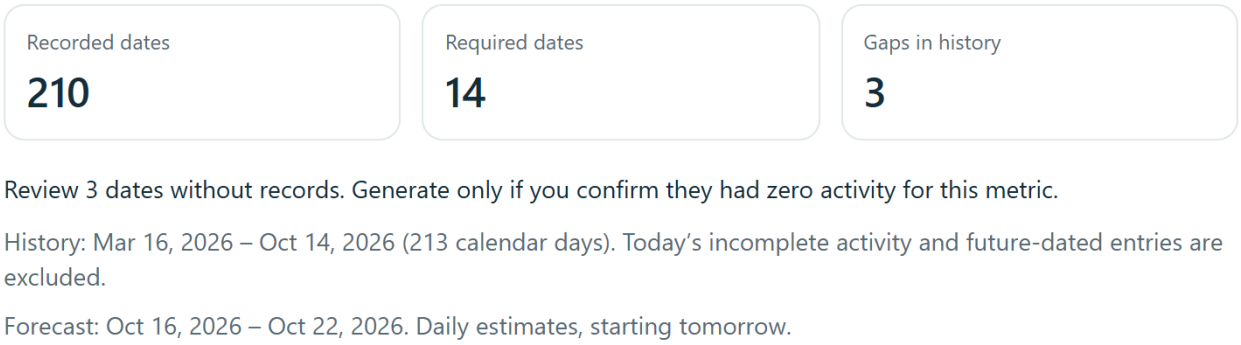


Figure 5.9: Forecast history review with recorded-date and gap counts.

The 7- and 30-day forecasts start tomorrow. Three-month forecasts sum daily predictions for the next three complete calendar months. The model projects the intervening bridge days but excludes them from the displayed horizon totals. Today's incomplete activity is never used as a completed training observation.

Holt's method updates a level and a trend from daily values [2]. In the equations below, y_t is the recorded or confirmed-zero amount, l_t is the level, b_t is the trend, and h is the number of days after the model origin. The initial level is the first value and the initial trend is the second value minus the first.

$$l_t = \alpha y_t + (1 - \alpha)(l_{t-1} + b_{t-1})$$

$$b_t = \beta(l_t - l_{t-1}) + (1 - \beta)b_{t-1}$$

$$\text{Forecast at } t + h = l_t + hb_t$$

Basira searches 121 alpha/beta combinations from 0 to 1 in increments of 0.1. Parameters are selected from the training portion only. The final 7, 30 or 90 observations form a chronological holdout, and a naive baseline repeats the last training value. The application reports mean absolute error and a clearly labeled nonzero-only MAPE; MAPE is not an accuracy percentage. Negative predictions are clamped to zero.

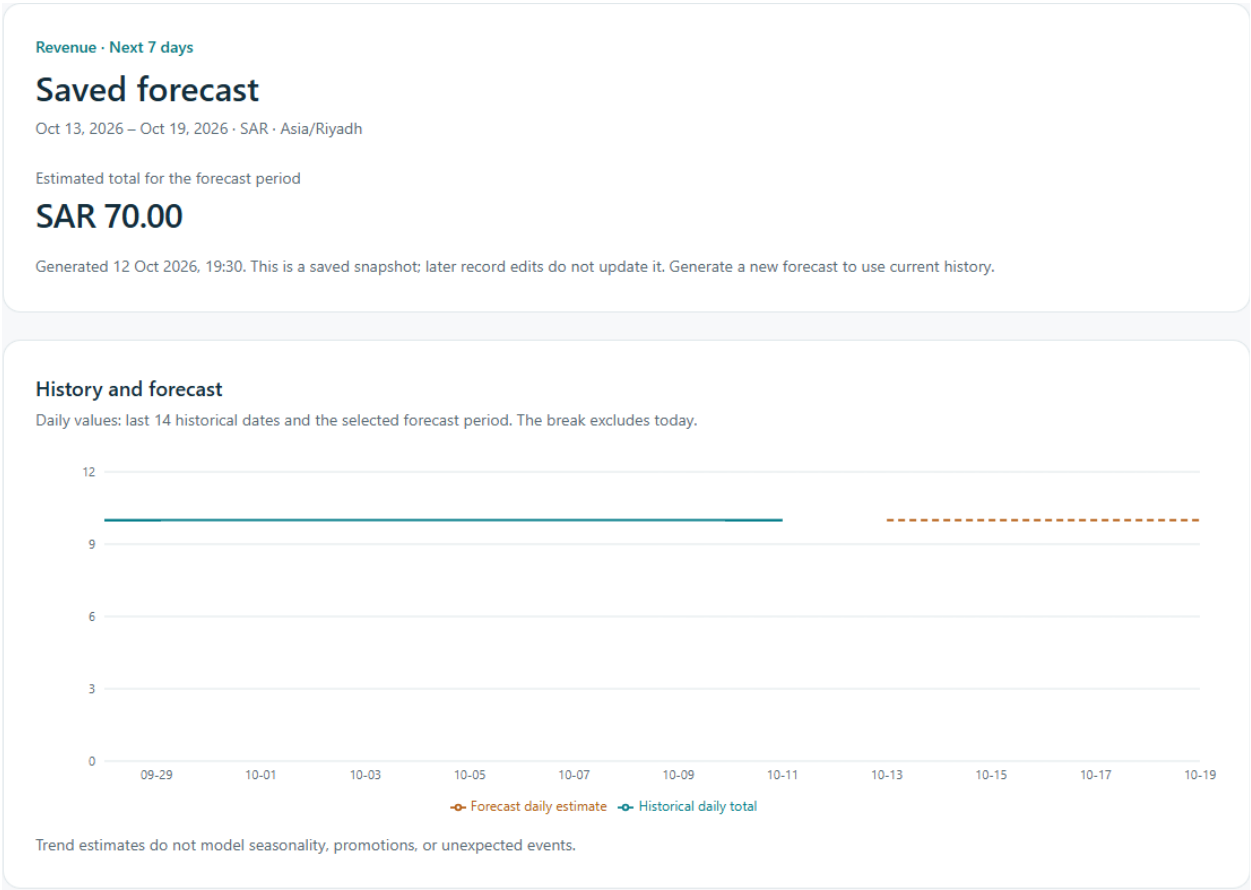


Figure 5.10: Saved forecast interface excerpt using a synthetic automated-test series.

ForecastRun preserves history, gap flags, parameters, evaluation values and algorithm version; ForecastPoint stores the future amounts. Later edits do not rewrite old results. Holt's method here has no seasonal component, causal variables or prediction intervals. A baseline warning and the stored errors support interpretation, but neither guarantees future accuracy.

5.2.12 Customer Feedback Analysis

The feedback module stores dated customer comments and displays their sentiment, main topic and short summary after analysis. New or imported feedback remains Unknown/Pending. The owner explicitly requests analysis, and the interface discloses that comment text is sent to the configured NVIDIA-hosted Gemma model [16]. Recording feedback alone does not send it to the provider.

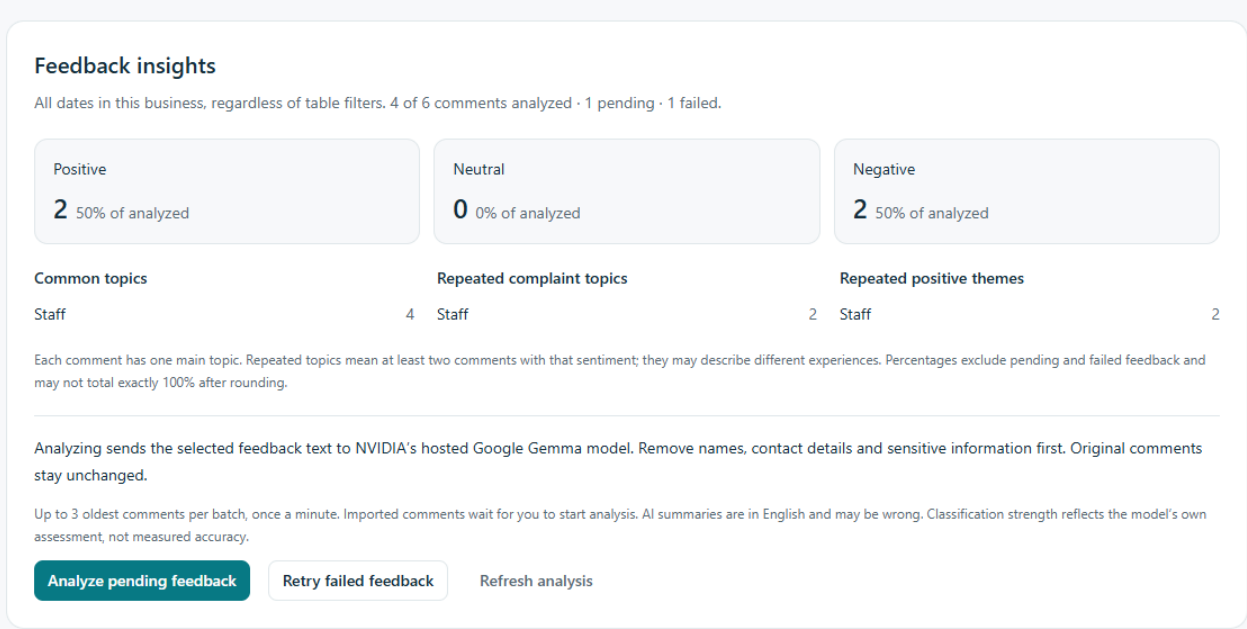


Figure 5.11: Feedback insights excerpt from an automated-test fixture with simulated classifications.

The server sends one JSON-encoded comment per request and expects sentiment, one topic, a summary and a numeric confidence value. Zod validates the returned structure, allowed values and length bounds independently of the provider response format. Invalid, truncated or refused output becomes a safe failure state; it is not displayed as a successful analysis. The model confidence represents its own classification strength and is not a measured or calibrated probability.

At most three comments are claimed in a batch. Provider calls run outside the database transaction with a 60-second deadline. A claim token, record version and expiring lease prevent a late response from overwriting edited feedback. Successful entries store provider, model, prompt version and completion time. Editing the feedback clears the old analysis.

Sentiment proportions use analyzed comments only, while pending and failed counts remain visible. The module supports Arabic input but requests English summaries for the current interface. Sarcasm, mixed opinions and dialect can still be misclassified. The fixture in Figure 5.11 illustrates the display and workflow; it is not evidence of live-model semantic accuracy.

5.2.13 AI Business Assistant

The business assistant explains a selected reporting period in plain language. A conversation fixes its dates, and each question receives a fresh server-built summary for those dates. The context contains calculated metrics, selected product/category aggregates, feedback counts and bounded saved forecast/alert summaries. The browser cannot supply replacement business figures or a different system prompt.

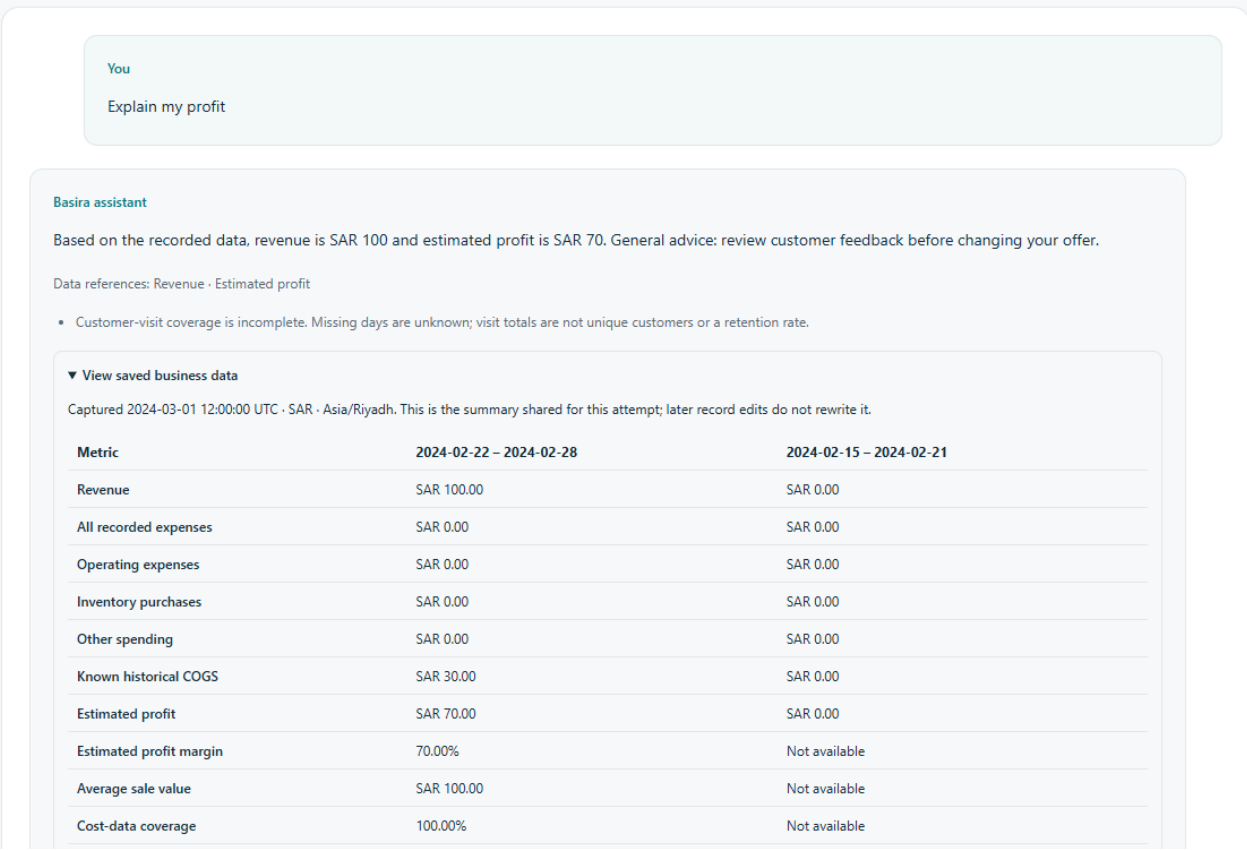


Figure 5.12: Assistant reply and saved context excerpt; reply supplied by an automated-test provider.

The request includes the question and at most three recent completed exchanges. Raw transactions, individual feedback and account credentials are excluded from the business-summary context. The model cannot browse, execute SQL, call tools or alter records. Replies follow a strict schema with an answer, in-scope flag, metric references and warnings. The application replaces a valid out-of-scope classification with its fixed business-only refusal.

ChatConversation and ChatMessage preserve completed answers and the context used for each attempt. Failed requests retain a retryable state. Request identifiers avoid duplicate question pairs, and claim/version checks prevent stale completions. Structural validation controls format, not truth: users must compare AI interpretations with the saved figures. General advice and possible causes must be distinguished from observations supported by the records.

5.2.14 Alerts and Notifications

Basira implements five configurable in-app alert rules. Each rule returns a clear state: Disabled, Insufficient data, Clear or Triggered. Thresholds are calculated locally from recorded values; the alert engine makes no AI request. Table 5.7 shows the default conditions.

Table 5.7: Default alert rules and data safeguards.

Rule	Default trigger and safeguard
Sales drop	Revenue decline of at least 20%; sales in both periods and a positive previous total are required.
Expense increase	All recorded spending increases by at least 20% against a positive previous total.
Low estimated margin	Estimated margin below 10%; positive revenue and complete historical cost coverage are required.
Negative feedback	Negative share above 40% among at least five analyzed comments.
Forecast revenue drop	Forecast daily average falls by more than 15% against its saved recent baseline; the run must be current.

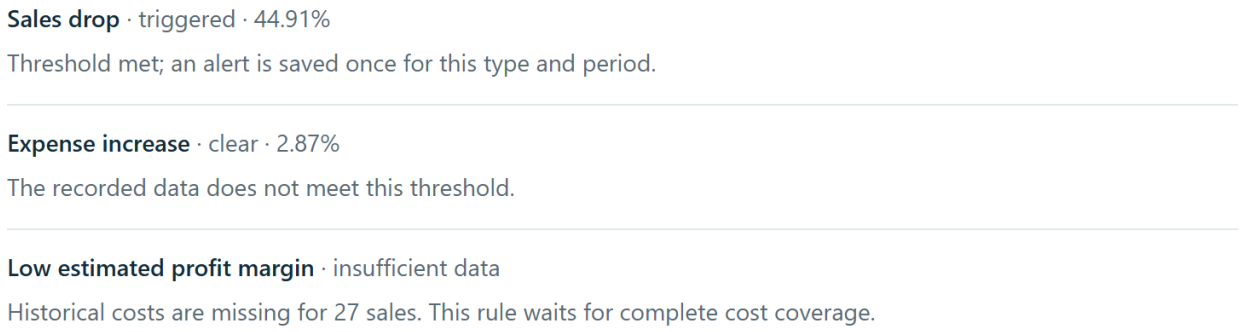
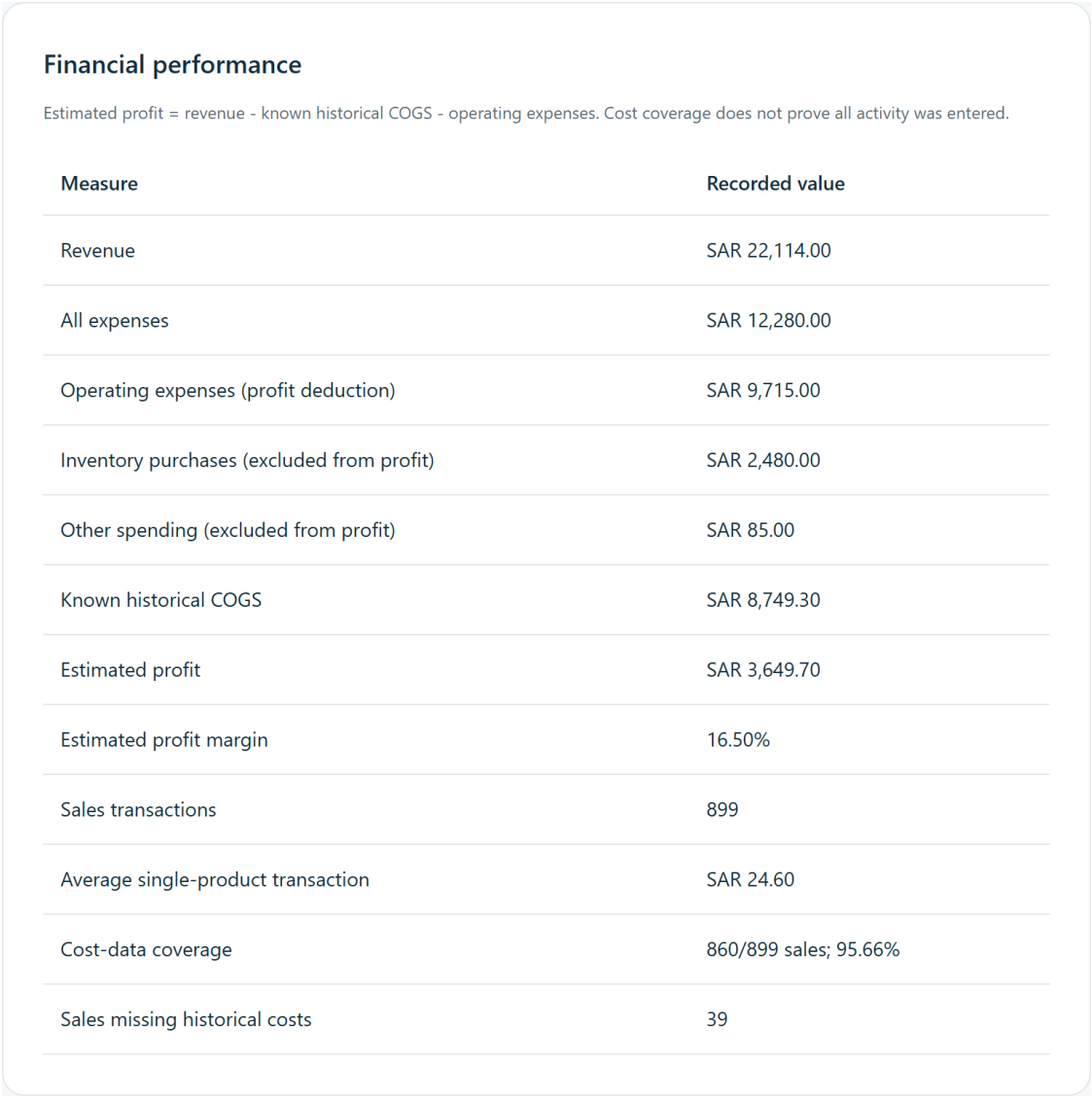


Figure 5.13: Alert-check excerpt for the synthetic cafe, including a cost-coverage warning.

Financial and feedback rules use the latest 30 completed days, with an equally long previous comparison where needed. Checks run after relevant writes and when the dashboard or alert workspace is loaded. Fingerprints prevent duplicate alerts for the same rule and evaluation period. Saved evidence remains historical after later record changes. Owners can mark alerts read or dismiss them; these actions do not resolve the business condition. No email delivery or background scheduler is implemented.

5.2.15 Report Generation

The Reports module creates weekly, monthly and custom business snapshots. Weekly and monthly choices use the previous completed week or month; custom periods are inclusive and limited to 366 days. Creation preserves the business labels, period, comparisons, KPIs, coverage, products, feedback aggregates and selected historical forecasts/alerts. Figure 5.14 shows part of a report generated from the synthetic cafe records.



Financial performance

Estimated profit = revenue - known historical COGS - operating expenses. Cost coverage does not prove all activity was entered.

Measure	Recorded value
Revenue	SAR 22,114.00
All expenses	SAR 12,280.00
Operating expenses (profit deduction)	SAR 9,715.00
Inventory purchases (excluded from profit)	SAR 2,480.00
Other spending (excluded from profit)	SAR 85.00
Known historical COGS	SAR 8,749.30
Estimated profit	SAR 3,649.70
Estimated profit margin	16.50%
Sales transactions	899
Average single-product transaction	SAR 24.60
Cost-data coverage	860/899 sales; 95.66%
Sales missing historical costs	39

Figure 5.14: Saved financial report for the synthetic cafe, 1 to 30 September 2026.

The report is immutable: later record edits do not change its figures. PDF downloads are generated from the saved snapshot using the local React PDF renderer [17], with no paid PDF service. A deterministic written summary is available without AI. The owner may separately request an AI executive summary; if the provider fails, the stored report and PDF remain available. Reports are generated on demand and are not automatically scheduled or emailed.

5.2.16 Administration Panel

The administrator area provides platform statistics, a searchable user list, confirmed activation/suspension and a read-only business directory. Figure 5.15 shows the statistics interface using synthetic test accounts. Administrative routes check the current active ADMIN role on the server; hiding a navigation item is not the access-control mechanism.

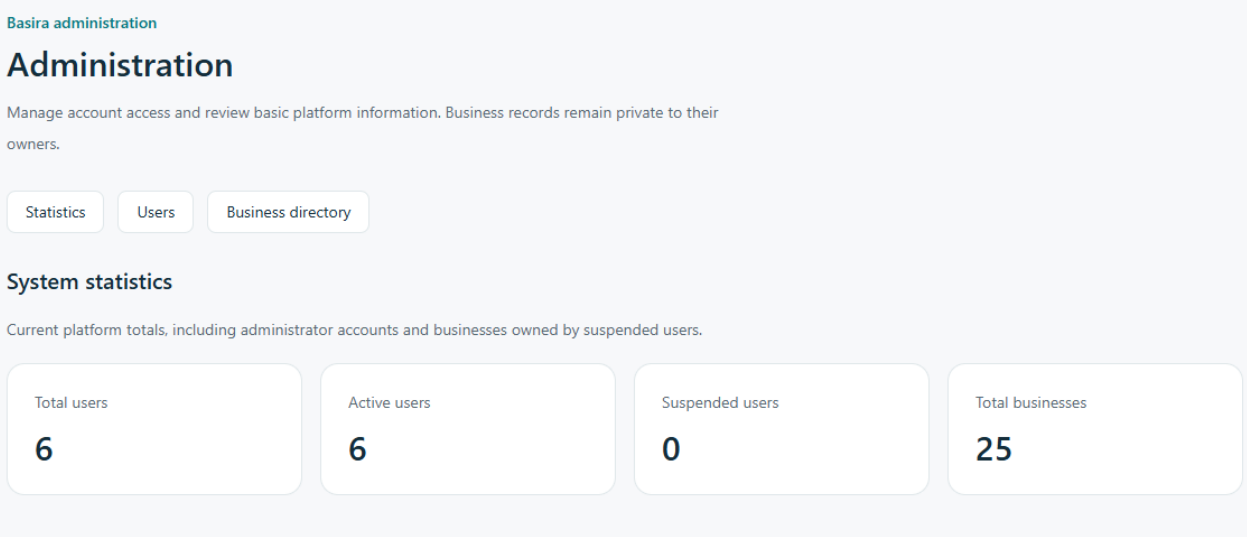


Figure 5.15: Administration statistics from an automated-test fixture.

Administrators receive limited account and business metadata, not other owners' sales, expenses, feedback, forecasts or conversations. Business-content services retain the ownership restriction even for an administrator. Status changes require confirmation and the observed status version. Self-suspension is blocked, and serialized status changes protect the supported workflow from suspending the last active administrator.

Suspension preserves business data while blocking subsequent protected requests. Reactivation restores access according to the session policy. The first administrator is granted through an operator-only command for a specifically selected existing account; there is no public role-promotion screen or default administrator password.

5.2.17 Security and Error Handling

Security is enforced at several boundaries: the HTTP handler checks authentication and request shape, the service verifies active ownership, and PostgreSQL constraints protect relationships and stored values. Business selection and client-supplied identifiers never replace those checks. Detailed test evidence for these controls is reserved for Chapter 6.

State-changing JSON routes check the request origin and impose bounded request sizes. Zod rejects unknown or malformed fields. File parsing applies upload and expansion limits before confirmation. Versions reject stale edits, while transactions and shared business-row locks coordinate imports, financial writes and permanent settings locks. Private API responses use no-store caching policies.

Database connection details, authentication secrets and provider keys remain server-side. Error responses use safe messages rather than database errors, stack traces or raw provider responses. AI timeout, rate-limit, configuration and malformed-output states remain distinguishable internally while the interface explains the action available to the user. Loading states prevent accidental repeat submissions, and retry identifiers prevent duplicate writes where the operation supports them.

The interface supports light/dark themes, responsive navigation, labeled form controls, keyboard interaction and table-based alternatives to charts. These implementation choices address usability and accessibility requirements without claiming that every assistive technology or production workload has been validated. Account permissions are deliberately separate from business-content ownership.

5.2.18 Local Execution and Deployment Preparation

The local environment uses separate development and test PostgreSQL databases. Dependency installation generates the Prisma client, and reviewed migrations establish the schema. The normal local launch sequence is npm ci, npm run db:deploy and npm run dev. A production build uses npm run build followed by npm run start. Development and test database credentials are configured outside source control.

An optional local-only seed creates the synthetic cafe and service-business demonstrations. It is idempotent and does not overwrite existing demo edits. It does not fabricate AI analysis or create a privileged administrator. Google and NVIDIA configuration are independent of the deterministic dashboard, forecasting, alert and report features.

Deployment preparation includes environment preflight checks, a database migration workflow, bounded connection pooling, a CI configuration and hosting instructions. A real deployment still requires the selected hosting account, production PostgreSQL, HTTPS origin, OAuth callback, secrets, backup arrangements and live-provider checks. Hosting has not been connected for this local submission, so public availability, hosted capacity and continuous operation are not claimed.

5.3 Chapter Summary

This chapter described the software and hardware used to implement Basira, the client-server communication path, ORM mapping and the main application modules. Shared ownership checks and historical snapshots connect data entry to analysis without silently changing past financial meaning. The implementation provides the prototype needed to examine the requirements through functional, integration and interface testing in Chapter 6. Forecasts and AI interpretations remain decision-support outputs whose usefulness depends on data quality and the limits explained in the interface.

References

[1] Power, D. J. (2002). Decision Support Systems: Concepts and Resources for Managers. Quorum Books.
Link: https://scholarworks.uni.edu/facbook/67/
Alternative preview:
https://books.google.com/books/about/Decision_Support_Systems.html?id=9NA6QMcte3cC

[2] Hyndman, R. J., & Athanasopoulos, G. (2021). Forecasting: Principles and Practice (3rd ed.). OTexts.
Link: https://otexts.com/fpp3/

[3] Jurafsky, D., & Martin, J. H. (2025). Speech and Language Processing (3rd ed. draft). Stanford University.
Link: https://web.stanford.edu/~jurafsky/slp3/

[4] Microsoft. (2026). Power BI pricing and licensing information.
Link: https://www.microsoft.com/en-us/power-platform/products/power-bi/pricing

[5] Zoho. (2026). Zoho Analytics pricing.
Link: https://www.zoho.com/analytics/pricing.html

[6] Zoho. (2026). Zoho Analytics dashboards documentation.
Link: https://www.zoho.com/analytics/help/dashboard/

[7] Intuit QuickBooks. (2026). QuickBooks Online pricing and product information.
Link: https://quickbooks.intuit.com/global/pricing/

[8] Oracle NetSuite. (2025). ERP Dashboards for Businesses.
Link: https://www.netsuite.com/portal/resource/articles/erp/erp-dashboard.shtml

[9] Beck, K., Beedle, M., van Bennekum, A., Cockburn, A., Cunningham, W., Fowler, M., Grenning, J., Highsmith, J., Hunt, A., Jeffries, R., Kern, J., Marick, B., Martin, R. C., Mellor, S., Schwaber, K., Sutherland, J., and Thomas, D. (2001). Manifesto for Agile Software Development.
Link: https://agilemanifesto.org/

[10] Beck, K., Beedle, M., van Bennekum, A., Cockburn, A., Cunningham, W., Fowler, M., Grenning, J., Highsmith, J., Hunt, A., Jeffries, R., Kern, J., Marick, B., Martin, R. C., Mellor, S., Schwaber, K., Sutherland, J., and Thomas, D. (2001). Principles behind the Agile Manifesto.
Link: https://agilemanifesto.org/principles.html

[11] Object Management Group. (2017). Unified Modeling Language Specification Version 2.5.1.
Link: https://www.omg.org/spec/UML/2.5.1/About-UML

[12] Google AI for Developers. (n.d.). Gemini API Documentation.
Link: https://ai.google.dev/gemini-api/docs

[13] Vercel. (2026). Route Handlers. Next.js documentation. Retrieved October 15, 2026.
Link: https://nextjs.org/docs/app/getting-started/route-handlers

[14] Prisma. (n.d.). Upgrade to Prisma ORM v7. Prisma documentation. Retrieved October 15, 2026.
Link: https://www.prisma.io/docs/guides/upgrade-prisma-orm/v7

[15] NextAuth.js. (n.d.). Options. NextAuth.js v4 documentation. Retrieved October 15, 2026.
Link: https://next-auth.js.org/configuration/options

[16] NVIDIA. (n.d.). Google / gemma-4-31b-it. NVIDIA API documentation. Retrieved October 15, 2026.
Link: https://docs.api.nvidia.com/nim/reference/google-gemma-4-31b-it

[17] React-pdf. (n.d.). Node API. React-pdf v4 documentation. Retrieved October 15, 2026.
Link: https://react-pdf.org/docs/v4/node